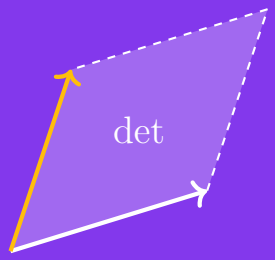
Linear Algebra, But Make It Make Sense

Lecture 3 – The determinant, two ways
(+ the start of eigenvalues)



Sasha Sutton – personal study notes, Fall 2026

My notes + proof sheet + fully solved exercises, remixed into one doc.

How to read this doc (the box legend)

DEFINITION	The lore. Learn these word for word, examiners love them.
THEOREM	The canon. Things that are always true.
RECEIPTS	Proofs. Every claim needs receipts.
EXAMPLE	Worked examples with actual numbers.
ML SIDE QUEST	Why a data scientist should care.
EXAM CHEAT CODE	"If they ask you *this*, you do *that*". Step-by-step recipes.
COMMON L	Classic mistakes. Don't be that person.
TL;DR	The summary for when you're cramming at 2am.

This lecture has two pathways

EVERYONE Section 1 (what a determinant *is*) and Sections 4 to 8.

PATH A: DEEP LORE Section 2: where the formula really comes from (permutations, signatures, Leibniz). For the curious and the proof-lovers.

PATH B: SPEEDRUN Section 3: how to *compute* any determinant fast (minors, cofactors, the $+-+-$ grid, row tricks, when it's 0, characteristic polynomials). For the exam and for next lectures.

You can do one path or both. They meet again in Section 4.

Everything here is real maths, just explained like a friend would. The vibes are optional, the rigour is not.

Contents

1 What a determinant IS (no jargon, promise) 3

1.1 The one-sentence definition . . . 3
1.2 The formulas you need on day one . . . 3
1.3 The picture: area, volume, orientation . . . 4
1.4 The rulebook (the "official" definition, made friendly) . . . 5
1.5 A bit of history . . . 6
1.6 ML side quests . . . 6
1.7 Choose your path . . . 7

2 Path A – Deep lore: where the formula comes from 8

2.1 The big idea: one entry per row and per column . . . 8
2.2 Permutations . . . 8
2.3 Transpositions: the simplest permutations . . . 9
2.4 The signature: the ± 1 of a permutation . . . 9
2.5 The Leibniz formula . . . 12
2.6 The rules, proved: multilinear and alternating . . . 13
2.7 Transpose and product . . . 14
2.8 Boss level: Laplace expansion from Leibniz . . . 14

3 Path B – Speedrun: how to compute any determinant 16

3.1 Small sizes: 1×1, 2×2, 3×3 . . . 16
3.2 Minors and cofactors . . . 16
3.3 The $+ - +-$ sign grid . . . 17
3.4 Cofactor (Laplace) expansion . . . 17
3.5 Row and column operations: the real power move . . . 19
3.6 When is a determinant zero? (read this carefully) . . . 20
3.7 Instant determinants (memorise these) . . . 22
3.8 Strategy: how to attack any determinant . . . 23
3.9 Toward the characteristic polynomial (why Path B matters next week) . . . 23
3.10 Bonus: cofactors give the inverse . . . 24

4 The properties everyone needs 25

4.1 Determinant and invertibility . . . 25
4.2 Determinant of a family of vectors, change of basis . . . 26
4.3 Determinant of an endomorphism . . . 27

5 Determinants of some particular matrices **27**

5.1 Orthogonal matrices: determinant ± 1 28

6 Eigenvalues, eigenvectors and the start of diagonalisation **29**

6.1 Definitions 29

6.2 Finding eigenvalues: this is where determinants come in 30

6.3 Diagonalisation 31

7 The exercises, fully solved **33**

8 Final boss: the cheat sheet **40**

8.1 "If they ask you this, do that" 41

8.2 Formula sheet 41

8.3 Hall of Ls 42

8.4 The whole lecture in 10 lines 42

1 What a determinant IS (no jargon, promise)

EVERYONE

Previously on Linear Algebra

- **Lecture 1:** vector spaces, subspaces, linear maps, rank, image, kernel.
- **Lecture 2:** matrices as linear maps, linear systems, range/rank/kernel of a matrix, inverse matrices, change of basis.
- **Today:** one single number that tells you whether a square matrix is invertible, how it scales space, and (next lectures) where its eigenvalues are. Then the beginning of diagonalisation.

1.1 The one-sentence definition

TL;DR // The determinant in plain words

The **determinant** of a square matrix A is **one number**, written $\det(A)$ or $|A|$, that answers:

"When I apply A to space, by what factor do areas (2D), volumes (3D), ... get multiplied, and does space get flipped?"

- $|\det A|$ = the **volume scaling factor**.
- The **sign** of $\det A$ = orientation: $+$ means "kept", $-$ means "mirrored".
- $\det A = 0$ means A **squashes** space into something flat (a line, a plane...). Information is destroyed, so A is **not invertible**.

That's it. Everything else in this lecture is either *how to compute* that number or *why* it behaves the way it does.

1.2 The formulas you need on day one

DEFINITION // Determinant of a 2×2 and a 3×3 matrix

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc, \qquad \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei + bfg + cdh - ceg - bdi - afh.$$

Vertical bars $|\cdots|$ mean "determinant of" (not absolute value!).

EXAMPLE // First computations

1. $\begin{vmatrix} 3 & 1 \\ 1 & 2 \end{vmatrix} = 3 \cdot 2 - 1 \cdot 1 = 5.$
2. $\begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} = 1 \cdot 1 - 2 \cdot 2 = -3.$ (Negative: orientation flipped.)
3. $\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = 4 - 4 = 0.$ (Second column = $2\times$ first: space gets squashed onto a line.)

4. $\begin{vmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 1 & 2 & 0 \end{vmatrix}$: here $a=2, b=1, c=3, d=0, e=-1, f=4, g=1, h=2, i=0$, so

$$\det = \underbrace{2\cdot(-1)\cdot 0}_{aei} + \underbrace{1\cdot 4\cdot 1}_{bfg} + \underbrace{3\cdot 0\cdot 2}_{cdh} - \underbrace{3\cdot(-1)\cdot 1}_{ceg} - \underbrace{1\cdot 0\cdot 0}_{bdi} - \underbrace{2\cdot 4\cdot 2}_{afh}$$

$$\det = 0+4+0+3-0-16=-9.$$

1.3 The picture: area, volume, orientation

Put the **columns** of a 2×2 matrix as two arrows v and w in the plane. They span a parallelogram.

THEOREM / PROPOSITION // Area of a parallelogram

For $v=\begin{pmatrix} a \\ c \end{pmatrix}$ and $w=\begin{pmatrix} b \\ d \end{pmatrix}$, the parallelogram spanned by v and w has area $\left|\det[\,v \mid w\,]\right| = |ad-bc|$. In 3D, the box (parallelepiped) spanned by v_1, v_2, v_3 has volume $\left|\det[\,v_1 \mid v_2 \mid v_3\,]\right|$.

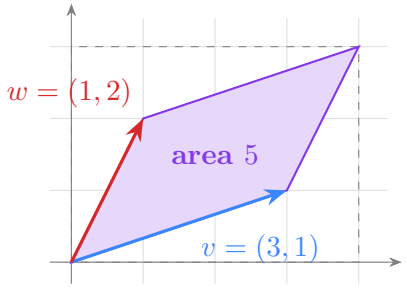
EXAMPLE // Checking the area formula with your hands

$v=(3,1)$, $w=(1,2)$, so $\det[\,v \mid w\,] = \begin{vmatrix} 3 & 1 \\ 1 & 2 \end{vmatrix} = 3\cdot 2 - 1\cdot 1 = 5$.

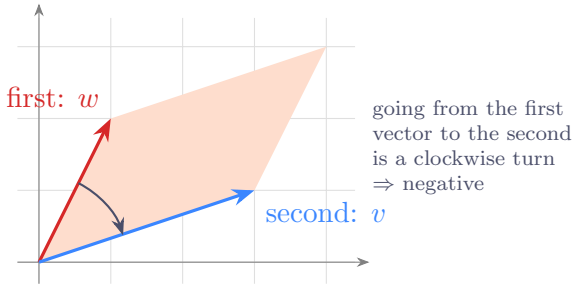
Now compute the area of the parallelogram the "primary school" way (see the picture): take the bounding rectangle and remove what's outside.

1. Bounding rectangle: $(3+1)\times(1+2) = 4\times 3 = 12$.
2. Two triangles with legs 3 and 1: $2\times\frac{1}{2}\cdot 3\cdot 1 = 3$.
3. Two triangles with legs 1 and 2: $2\times\frac{1}{2}\cdot 1\cdot 2 = 2$.
4. Two small 1×1 corner squares: $2\times 1 = 2$.
5. Area $= 12-3-2-2=5$. ✓ Same as the determinant.

With letters instead of numbers: $(a+b)(c+d) - ac - bd - 2bc = ad - bc$. That's literally where the 2×2 formula comes from.



$\det[\,v \mid w\,] = +5$

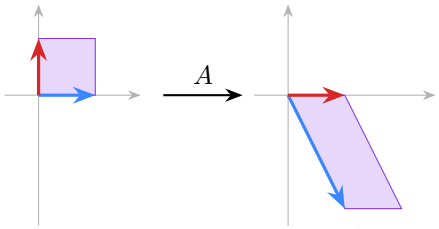


$\det[\,w \mid v\,] = -5$

Same parallelogram, same area 5. Going from the first vector to the second *counter-clockwise* gives $+$; *clockwise* gives $-$. **Orientation matters**: swapping the two columns flips the sign.

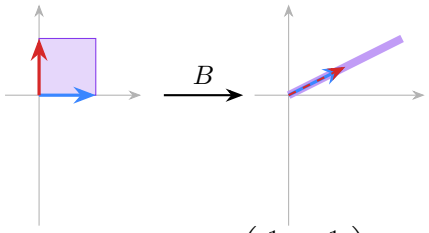
THEOREM / PROPOSITION // A matrix multiplies every area by $|\det A|$

If $A \in \mathbb{R}^{2\times 2}$, the linear map $x \mapsto Ax$ transforms any shape of area S into a shape of area $|\det A| \cdot S$. In $\mathbb{R}^d$: new volume $= |\det A| \times$ original volume.

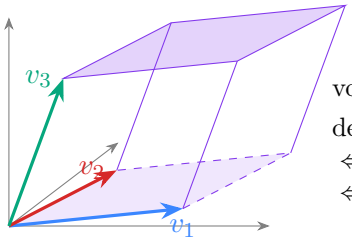


unit square
area 1

$A = \begin{pmatrix} 1 & 1 \\ -2 & 0 \end{pmatrix}$, $\det A = 2$
area 2



$B = \begin{pmatrix} 1 & 1 \\ 0.5 & 0.5 \end{pmatrix}$, $\det B = 0$
area 0: squashed flat



volume of the box $= |\det[\, v_1 \mid v_2 \mid v_3 \,]|$
$\det = 0 \iff$ the box is flat
$\iff v_1, v_2, v_3$ lie in a common plane
$\iff$ they are linearly dependent.

1.4 The rulebook (the "official" definition, made friendly)

There are several equivalent ways to define the determinant: (1) the **Leibniz formula** with permutations (Path A); (2) the **geometric** one (signed volume, above); (3) as **the unique function satisfying three rules**. The third one is the cleanest. Think of det as a function of the n columns $C_1, \dots, C_n$ of the matrix.

DEFINITION // The three rules that define det

det is the **only** function of n columns in $\mathbb{R}^n$ such that:

(R1) Normalisation: $\det(I_n) = 1$. *(the unit cube has volume 1)*

(R2) Alternating: swapping two columns flips the sign. *(orientation)*

(R3) Linear in each column separately (the other columns being fixed):

$$\det(\dots, \lambda u + \mu u', \dots) = \lambda \det(\dots, u, \dots) + \mu \det(\dots, u', \dots).$$

(stretch one side of a box by λ, the volume is multiplied by λ)

The same three rules hold for **rows** (because $\det A^\top = \det A$, see Section 4).

EXAMPLE // Seeing the rules on 2×2 matrices

1. **Swap the columns:** $\begin{vmatrix} b & a \\ d & c \end{vmatrix} = bc - ad = -(ad - bc) = -\begin{vmatrix} a & b \\ c & d \end{vmatrix}$. ✓ (R2)
2. **Scale one column by λ:** $\begin{vmatrix} \lambda a & b \\ \lambda c & d \end{vmatrix} = \lambda ad - \lambda bc = \lambda \begin{vmatrix} a & b \\ c & d \end{vmatrix}$. ✓ (R3)
3. **Add a vector to the first row:** $\begin{vmatrix} a+a' & b+b' \\ c & d \end{vmatrix} = (a+a')d - (b+b')c = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix}$.

✓ (R3, for rows)

THEOREM / PROPOSITION // Free consequences of the rules

(C1) Two equal columns $\Rightarrow \det = 0$.

(C2) A zero column $\Rightarrow \det = 0$.

(C3) Adding a multiple of one column to *another* column does **not** change det.

(C4) $\det(\lambda A) = \lambda^n \det(A)$ for $A \in \mathbb{R}^{n\times n}$.

RECEIPTS (proof) // from the three rules only

(C1) Swapping the two equal columns changes nothing, but (R2) says it flips the sign: $\det = -\det$, so $\det = 0$. *(C2)* The zero column is $0\cdot$(zero column); by (R3) we can pull the factor 0 out: $\det = 0 \cdot \det(\dots) = 0$. *(C3)* By (R3) in column j: $\det(\dots, C_j + \lambda C_i, \dots) = \det(\dots, C_j, \dots) + \lambda \det(\dots, C_i, \dots, C_i, \dots)$, and the last determinant has two equal columns, so it is 0 by (C1). *(C4)* λA has *every* one of its n columns multiplied by λ; pull out one factor λ per column. ■ *(and that's on periodt)*

COMMON L (don't do this) // The two classic traps

1. $\det(\lambda A) = \lambda^{\mathbf{n}} \det A$, **not** $\lambda \det A$. Example: $\det(2I_3) = 2^3 = 8$ (doubling every side of a cube multiplies its volume by 8).
2. $\det(A + B) \neq \det A + \det B$ in general. Example: $\det(I_2 + I_2) = \det(2I_2) = 4$ but $\det I_2 + \det I_2 = 2$. The determinant is linear in *each column separately*, not in the whole matrix. (Exercise 5 is built on this.)

1.5 A bit of history

Determinants came *before* matrices: they were introduced to decide whether a system of linear equations (as many unknowns as equations) has a unique solution. Cardano (2 equations, 2 unknowns, 1545), Leibniz (3×3, 1678), Maclaurin (4×4, 1748), Cramer (general formula, no proof), then Bézout, Lagrange, Gauss, Cauchy…

1.6 ML side quests

ML SIDE QUEST // Determinant as volume scaling

A linear map $A : \mathbb{R}^d \to \mathbb{R}^d$ turns the unit ball into a (hyper-)ellipsoid, and volumes are multiplied by $|\det A|$:

- $|\det A| > 1$: volumes are **expanded**;
- $0 < |\det A| < 1$: volumes are **contracted**;
- $\det A = 0$: some directions are **collapsed**.

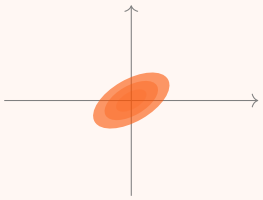
In ML, linear feature transformations can stretch, compress or collapse regions of the feature space. If $\det A \neq 0$ the original features can be recovered from $z = Ax$; if $\det A = 0$, some information is lost for good.

ML SIDE QUEST // Determinant in Gaussian models

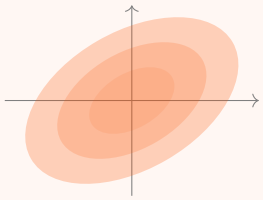
The density of a multivariate Gaussian $x \sim \mathcal{N}(\mu, \Sigma)$ in $\mathbb{R}^d$ is

$$p(x) = \frac{1}{(2\pi)^{d/2} \det(\Sigma)^{1/2}} \exp\Big(-\tfrac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu) \Big).$$

$\det(\Sigma)$ measures the **volume of the uncertainty ellipsoid**: large $\det \Sigma$ = spread-out distribution, small $\det \Sigma$ = concentrated one. The factor $\det(\Sigma)^{-1/2}$ is exactly what makes the total probability equal to 1: a bigger ellipsoid must have a lower density.

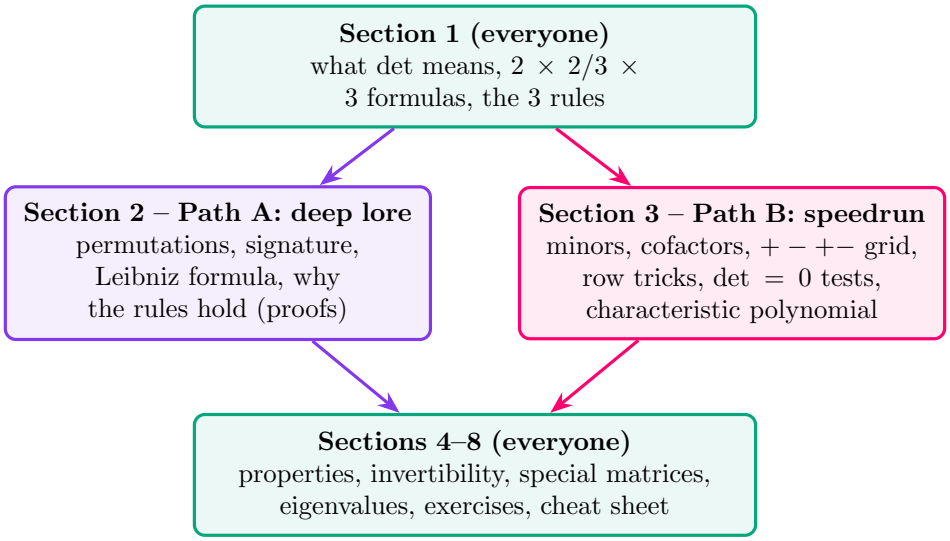


small $\det \Sigma$: concentrated



large $\det \Sigma$: spread out

1.7 Choose your path



TL;DR // Which path should I take?

- You want to pass the exam and use determinants for eigenvalues next week: **Path B** is enough.
- You want to understand *why* the formula looks like that, or you're aiming at research/PhD-level maths: do **Path A** too. It also contains the proofs of the rules Path B uses.

2 Path A – Deep lore: where the formula comes from

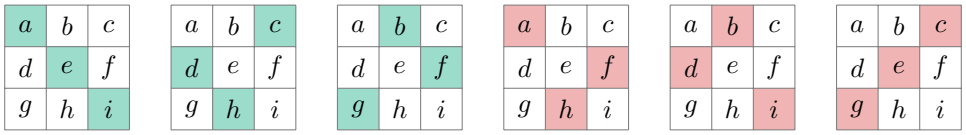
PATH A: DEEP LORE

TL;DR // Path A in one breath

A determinant is a **sum over all the ways to pick one entry in each row and each column**. Each pick gives a product of n numbers, weighted by $+1$ or -1. The "ways to pick" are **permutations**; the ± 1 is their **signature**. From that single formula (Leibniz), every rule of Section 1 can be proved.

2.1 The big idea: one entry per row and per column

Look again at the 3×3 formula: $aei + bfg + cdh - ceg - bdi - afh$. Every term is a product of 3 entries taken from **3 different rows and 3 different columns**, like placing 3 rooks on a 3×3 chessboard so that none attacks another.



The 6 ways to choose one entry per row and per column in a 3×3 matrix. Green = counted with $+$, red = counted with $-$. Exactly the 6 terms of the formula.

Two questions remain: **how do we list all the choices** for an $n \times n$ matrix, and **which sign** does each one get? Answers: permutations and their signature.

2.2 Permutations

DEFINITION // Permutation, $\mathfrak{S}_n$

Let $n \in \mathbb{N}^*$. A **permutation** of $[\![1, n]\!] = \{1, \dots, n\}$ is a bijection $\sigma : [\![1, n]\!] \to [\![1, n]\!]$. The set of all permutations is written $\mathfrak{S}_n$.
One-line notation: we write $\sigma = [\sigma(1)\ \sigma(2)\ \cdots\ \sigma(n)]$. Example: $\sigma = [3\ 1\ 2]$ means $\sigma(1) = 3$, $\sigma(2) = 1$, $\sigma(3) = 2$.

Link with the chessboard: choosing, in each column j, the entry of row $\sigma(j)$ gives one entry per column; it gives one entry per row exactly when σ is a bijection.

THEOREM / PROPOSITION // Counting permutations

$\mathrm{Card}(\mathfrak{S}_n) = n!$.

RECEIPTS (proof) //

To build σ, choose $\sigma(1)$: n possibilities. Then $\sigma(2)$ must differ from $\sigma(1)$ (injectivity): $n - 1$ possibilities. Then $n - 2$ for $\sigma(3)$, and so on down to 1. Total $n(n-1)\cdots 2 \cdot 1 = n!$, and each sequence of choices gives exactly one permutation. ■ *(and that's on periodt)*

EXAMPLE // $\mathfrak{S}_3$, all six of them

$[1\,2\,3]$ (identity), $[1\,3\,2]$, $[2\,1\,3]$, $[2\,3\,1]$, $[3\,1\,2]$, $[3\,2\,1]$. That's $3! = 6$. For $n = 4$ there are already 24, for $n = 10$ there are 3 628 800.

2.3 Transpositions: the simplest permutations

DEFINITION // Transposition

For $i \neq j$ in $[\![1, n]\!]$, the **transposition** $\tau_{i,j}$ swaps i and j and fixes everything else:

$$\tau_{i,j}(i) = j, \qquad \tau_{i,j}(j) = i, \qquad \tau_{i,j}(k) = k \quad \text{for } k \notin \{i, j\}.$$

(Convention: $\tau_{i,i} = \mathrm{Id}$.) A transposition is its own inverse: $\tau_{i,j} \circ \tau_{i,j} = \mathrm{Id}$ (swap twice = do nothing).

THEOREM / PROPOSITION // Every permutation is a product of transpositions

For every $\sigma \in \mathfrak{S}_n$ there exist transpositions $\tau_1, \ldots, \tau_T$ with $\sigma = \tau_1 \circ \tau_2 \circ \cdots \circ \tau_T$.

RECEIPTS (proof) // induction on n

Base case $n = 1$: $\mathfrak{S}_1 = \{\mathrm{Id}\}$, a composition of zero transpositions.
Induction step. Assume the result for $n - 1$ and let $\sigma \in \mathfrak{S}_n$, $k = \sigma(n)$.

- If $k = n$: σ fixes n, so it is really a permutation of $\{1, \ldots, n-1\}$, a product of transpositions by the induction hypothesis.
- If $k \neq n$: let $\rho = \tau_{k,n} \circ \sigma$. Then $\rho(n) = \tau_{k,n}(k) = n$, so ρ fixes n and is a product of transpositions by the first case. Since $\tau_{k,n}^{-1} = \tau_{k,n}$, $\sigma = \tau_{k,n} \circ \rho$ is a product of transpositions too. ■ *(and that's on periodt)*

EXAMPLE // Decomposing $\sigma = [3\ 1\ 2]$

Claim: $\sigma = \tau_{1,3} \circ \tau_{2,3}$ (apply $\tau_{2,3}$ first, then $\tau_{1,3}$). Check each input:

1. $1 \xrightarrow{\tau_{2,3}} 1 \xrightarrow{\tau_{1,3}} 3 = \sigma(1)$ ✓
2. $2 \xrightarrow{\tau_{2,3}} 3 \xrightarrow{\tau_{1,3}} 1 = \sigma(2)$ ✓
3. $3 \xrightarrow{\tau_{2,3}} 2 \xrightarrow{\tau_{1,3}} 2 = \sigma(3)$ ✓

Two transpositions. (Other decompositions exist, but they *always* use an even number: that's the next theorem.)

2.4 The signature: the ± 1 of a permutation

DEFINITION // Signature, inversion

The **signature** of $\sigma \in \mathfrak{S}_n$ is

$$\varepsilon(\sigma) = \prod_{1 \leq i < j \leq n} \frac{\sigma(j) - \sigma(i)}{j - i}.$$

An **inversion** of σ is a pair (i, j) with $i < j$ and $\sigma(i) > \sigma(j)$ (a pair that σ puts "in the wrong order"). $N(\sigma)$ denotes the number of inversions.

EXAMPLE // Signature of [3 1 2], two ways

With the product formula (pairs $i < j$):

1. $(1,2)$: $\dfrac{\sigma(2)-\sigma(1)}{2-1} = \dfrac{1-3}{1} = -2.$
2. $(1,3)$: $\dfrac{\sigma(3)-\sigma(1)}{3-1} = \dfrac{2-3}{2} = -\dfrac{1}{2}.$
3. $(2,3)$: $\dfrac{\sigma(3)-\sigma(2)}{3-2} = \dfrac{2-1}{1} = 1.$
4. Product: $(-2)\cdot(-\frac{1}{2})\cdot 1 = +1.$

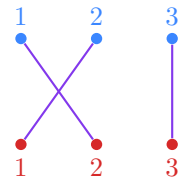
By counting inversions: $(1,2)$: $3 > 1$ inversion; $(1,3)$: $3 > 2$ inversion; $(2,3)$: $1 < 2$ fine. $N(\sigma) = 2$, so $(-1)^2 = +1$. Same answer, much faster.

THEOREM / PROPOSITION // Signature = parity of the number of inversions

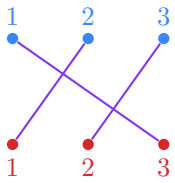
For every $\sigma \in \mathfrak{S}_n$: $\varepsilon(\sigma) = (-1)^{N(\sigma)}$. In particular $\varepsilon(\sigma) \in \{-1, +1\}$.

RECEIPTS (proof) //

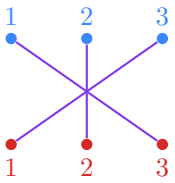
The denominator $\prod_{i<j}(j-i)$ is a product of positive numbers. In the numerator $\prod_{i<j}(\sigma(j)-\sigma(i))$, since σ is a bijection, the unordered pairs $\{\sigma(i),\sigma(j)\}$ run exactly once through all unordered pairs of distinct elements of $[\![1,n]\!]$. So, *up to sign*, numerator and denominator are the same product: $|\varepsilon(\sigma)| = 1$. The factor $\sigma(j)-\sigma(i)$ is negative exactly when (i,j) is an inversion, so the numerator has exactly $N(\sigma)$ negative factors: $\varepsilon(\sigma) = (-1)^{N(\sigma)}$. ■ *(and that's on periodt)*



[2 1 3]: 1 crossing
$\varepsilon = -1$



[3 1 2]: 2 crossings
$\varepsilon = +1$



[3 2 1]: 3 crossings
$\varepsilon = -1$

top: positions j
bottom: values $\sigma(j)$
crossings = inversions

THEOREM / PROPOSITION // The signature is multiplicative

For all $\sigma, \pi \in \mathfrak{S}_n$: $\varepsilon(\sigma \circ \pi) = \varepsilon(\sigma)\,\varepsilon(\pi)$.

RECEIPTS (proof) // telescoping the product

The quotient $\frac{\sigma(j)-\sigma(i)}{j-i}$ doesn't change if we swap i and j, so $\varepsilon(\sigma)$ is a product over *unordered* pairs $\{i,j\}$. Multiply and divide each factor by $\pi(j)-\pi(i) \neq 0$:

$$\varepsilon(\sigma\circ\pi) = \prod_{\{i,j\}} \frac{\sigma(\pi(j))-\sigma(\pi(i))}{\pi(j)-\pi(i)} \cdot \prod_{\{i,j\}} \frac{\pi(j)-\pi(i)}{j-i}.$$

The second product is $\varepsilon(\pi)$. In the first, put $k = \pi(i)$, $l = \pi(j)$: as $\{i,j\}$ runs over all unordered pairs, so does $\{k,l\}$ (π is a bijection). So the first product is $\prod_{\{k,l\}} \frac{\sigma(l)-\sigma(k)}{l-k} = \varepsilon(\sigma)$. ■ *(and that's on periodt)*

THEOREM / PROPOSITION // A transposition has signature -1

For $i < j$, $\tau_{i,j}$ has exactly $2(j-i-1)+1$ inversions, an odd number, so $\varepsilon(\tau_{i,j}) = -1$.

RECEIPTS (proof) // count the inversions

Write $\tau = \tau_{i,j}$. A pair not involving i or j is untouched: no inversion. The inversions are:

- (i,k) with $i < k < j$: $\tau(i) = j > k = \tau(k)$. That's $j-i-1$ inversions.
- (k,j) with $i < k < j$: $\tau(k) = k > i = \tau(j)$. Another $j-i-1$.
- (i,j) itself: $\tau(i) = j > i = \tau(j)$. One more.

(Pairs (k,i) with $k < i$ or (j,k) with $k > j$ are not inversions.) Total $2(j-i-1)+1$: odd. ■ *(and that's on periodt)*

THEOREM / PROPOSITION // Consequences

1. If $\sigma = \tau_1 \circ \cdots \circ \tau_T$ (transpositions), then $\varepsilon(\sigma) = (-1)^T$. So the *parity* of T never depends on the chosen decomposition.
2. $\varepsilon(\sigma^{-1}) = \varepsilon(\sigma)$.
3. $\varepsilon(\tau \circ \sigma) = -\varepsilon(\sigma)$ for any transposition τ.

RECEIPTS (proof) //

(1) Multiplicativity: $\varepsilon(\sigma) = \varepsilon(\tau_1)\cdots\varepsilon(\tau_T) = (-1)^T$. (2) $1 = \varepsilon(\mathrm{Id}) = \varepsilon(\sigma \circ \sigma^{-1}) = \varepsilon(\sigma)\varepsilon(\sigma^{-1})$, and both factors are ± 1, so they're equal. (3) Multiplicativity and $\varepsilon(\tau) = -1$. ■ *(and that's on periodt)*

$\sigma \in \mathfrak{S}_3$	inversions	$N(\sigma)$	$\varepsilon(\sigma)$
[1 2 3]	none	0	$+1$
[1 3 2]	$(2,3)$	1	-1
[2 1 3]	$(1,2)$	1	-1
[2 3 1]	$(1,3),(2,3)$	2	$+1$
[3 1 2]	$(1,2),(1,3)$	2	$+1$
[3 2 1]	$(1,2),(1,3),(2,3)$	3	-1

EXAM CHEAT CODE // "Compute the signature of σ"

1. Write σ in one-line notation $[\sigma(1)\ \cdots\ \sigma(n)]$.
2. For each entry, count how many entries **to its right are smaller**. Add these counts: that's $N(\sigma)$.
3. $\varepsilon(\sigma) = +1$ if $N(\sigma)$ is even, -1 if odd.

Example: [4 1 3 2]: after 4: three smaller $(1,3,2)$; after 1: none; after 3: one (2); after 2: none. $N = 4$, so $\varepsilon = +1$.

2.5 The Leibniz formula

DEFINITION // Determinant (Leibniz formula)

For $A = (a_{ij}) \in \mathcal{M}_n(\mathbb{R})$:

$$\det(A) = \sum_{\sigma \in \mathfrak{S}_n} \varepsilon(\sigma) \prod_{j=1}^{n} a_{\sigma(j),j}$$

In words: for each permutation σ, take in each column j the entry of row $\sigma(j)$, multiply these n entries, multiply by the signature, and add over all $n!$ permutations.

EXAMPLE // Recovering the 2×2 formula

$\mathfrak{S}_2 = \{[1\,2], [2\,1]\}$ with signatures $+1$ and -1.

1. $\sigma = [1\,2]$: $a_{11}a_{22}$, sign $+$.
2. $\sigma = [2\,1]$: $a_{21}a_{12}$, sign $-$.

$\det A = a_{11}a_{22} - a_{21}a_{12}$, i.e. $ad - bc$ for $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. ✓

EXAMPLE // Recovering the 3×3 formula

$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$, so $a_{11} = a$, $a_{12} = b$, …, $a_{33} = i$. Using the table of $\mathfrak{S}_3$ above:

σ	$a_{\sigma(1),1}\, a_{\sigma(2),2}\, a_{\sigma(3),3}$	$\varepsilon(\sigma)$	term
$[1\,2\,3]$	$a_{11}a_{22}a_{33} = aei$	$+1$	$+aei$
$[1\,3\,2]$	$a_{11}a_{32}a_{23} = ahf$	-1	$-afh$
$[2\,1\,3]$	$a_{21}a_{12}a_{33} = dbi$	-1	$-bdi$
$[2\,3\,1]$	$a_{21}a_{32}a_{13} = dhc$	$+1$	$+cdh$
$[3\,1\,2]$	$a_{31}a_{12}a_{23} = gbf$	$+1$	$+bfg$
$[3\,2\,1]$	$a_{31}a_{22}a_{13} = gec$	-1	$-ceg$

Sum: $aei + bfg + cdh - ceg - bdi - afh$. ✓ The mysterious formula is just Leibniz for $n = 3$.

THEOREM / PROPOSITION // Normalisation

For every $n \geq 1$, $\det(I_n) = 1$.

RECEIPTS (proof) //

In $\prod_j (I_n)_{\sigma(j),j}$, a factor is 0 unless $\sigma(j) = j$. So only $\sigma = \mathrm{Id}$ survives, with product 1 and $\varepsilon(\mathrm{Id}) = +1$. ■ *(and that's on periodt)*

COMMON L (don't do this) // Never compute a big determinant with Leibniz

The formula has $n!$ terms, each a product of n numbers:

n	2	3	4	5	10	20
terms $n!$	2	6	24	120	3 628 800	$\approx 2.4 \times 10^{18}$

Leibniz is for *understanding* and *proving*. To *compute*, use row operations (about n^3 operations) or cofactor expansion on a row full of zeros: that's Path B.

2.6 The rules, proved: multilinear and alternating

DEFINITION // Multilinear form, alternating form

Let E be a vector space over $\mathbb{K}$ ($\mathbb{R}$ or $\mathbb{C}$). A map $f : E^n \to \mathbb{K}$ is a **multilinear form** if it is linear in each argument separately: for each i and fixed $u_1, \dots, u_{i-1}, u_{i+1}, \dots, u_n$, the map $x \mapsto f(u_1, \dots, u_{i-1}, x, u_{i+1}, \dots, u_n)$ is linear.
f is **alternating** if $f(u_1, \dots, u_n) = 0$ whenever two arguments are equal ($u_i = u_j$ for some $i \neq j$).

THEOREM / PROPOSITION // The determinant is multilinear in the columns

For column vectors $u, v \in \mathbb{K}^n$ and scalars α, β: $\det(C_1, \dots, \alpha u + \beta v, \dots, C_n) = \alpha \det(C_1, \dots, u, \dots, C_n) + \beta \det(C_1, \dots, v, \dots, C_n)$.

RECEIPTS (proof) //

In each Leibniz term $\varepsilon(\sigma) \prod_k a_{\sigma(k),k}$, exactly one factor comes from column j: it is $\alpha u_{\sigma(j)} + \beta v_{\sigma(j)}$. Expanding this sum splits every term in two, hence the whole sum into $\alpha \det(\dots, u, \dots) + \beta \det(\dots, v, \dots)$. ■ *(and that's on periodt)*

THEOREM / PROPOSITION // Swapping columns

Exchanging two columns of a square matrix multiplies its determinant by -1. Consequently, if two columns are equal, the determinant is 0.

RECEIPTS (proof) // reindex the sum

Let B be A with columns p and q swapped, and $\tau = \tau_{p,q}$. Then $b_{i,j} = a_{i,\tau(j)}$. In Leibniz for B, substitute $\sigma = \sigma' \circ \tau$ (as σ' runs over $\mathfrak{S}_n$, so does σ): the product $\prod_j b_{\sigma(j),j} = \prod_j a_{\sigma'(\tau(j)),\tau(j)} = \prod_k a_{\sigma'(k),k}$ (reindex $k = \tau(j)$), while $\varepsilon(\sigma' \circ \tau) = -\varepsilon(\sigma')$. Hence $\det B = -\det A$. If columns p and q are equal, swapping them gives back A, so $\det A = -\det A$, i.e. $\det A = 0$. ■ *(and that's on periodt)*

THEOREM / PROPOSITION // Alternating ⇒ sign change

If f is multilinear and alternating, exchanging two arguments multiplies f by -1.

RECEIPTS (proof) // the $(u+v, u+v)$ trick

Only the two positions matter. By alternation and multilinearity: $0 = f(\dots, u+v, \dots, u+v, \dots) = \underbrace{f(\dots, u, \dots, u, \dots)}_{0} + f(\dots, u, \dots, v, \dots) + f(\dots, v, \dots, u, \dots) + \underbrace{f(\dots, v, \dots, v, \dots)}_{0}$.
So $f(\dots, v, \dots, u, \dots) = -f(\dots, u, \dots, v, \dots)$. ■ *(and that's on periodt)*

THEOREM / PROPOSITION // Characterisation: the rulebook really defines det

Let E have dimension n and basis $\mathcal{B} = (e_1, \dots, e_n)$. There is a **unique** map $\det_{\mathcal{B}} : E^n \to \mathbb{K}$ that is (i) multilinear, (ii) alternating, (iii) $\det_{\mathcal{B}}(e_1, \dots, e_n) = 1$. If X is the matrix whose j-th

column is $[v_j]_{\mathcal{B}}$, then $\det_{\mathcal{B}}(v_1, \ldots, v_n) = \det(X)$.

RECEIPTS (proof) // existence + uniqueness

Existence. Define $\det_{\mathcal{B}}(v_1, \ldots, v_n) = \det(X)$. Coordinates depend linearly on vectors, and the matrix determinant is multilinear and alternating in its columns (just proved), so $\det_{\mathcal{B}}$ is too. And $(e_1, \ldots, e_n)$ has coordinate matrix I_n, of determinant 1.
Uniqueness. Let f satisfy (i)–(iii). Write $v_j = \sum_{i=1}^{n} a_{ij} e_i$. Expanding by multilinearity in each argument:

$$f(v_1, \ldots, v_n) = \sum_{i_1=1}^{n} \cdots \sum_{i_n=1}^{n} a_{i_1 1} \cdots a_{i_n n} \, f(e_{i_1}, \ldots, e_{i_n}).$$

If two indices i_k are equal, $f(e_{i_1}, \ldots, e_{i_n}) = 0$ (alternating). So only tuples using each index exactly once survive, i.e. $(i_1, \ldots, i_n) = (\sigma(1), \ldots, \sigma(n))$ with $\sigma \in \mathfrak{S}_n$. Sorting $e_{\sigma(1)}, \ldots, e_{\sigma(n)}$ back into order with transpositions flips the sign once per transposition, so $f(e_{\sigma(1)}, \ldots, e_{\sigma(n)}) = \varepsilon(\sigma) f(e_1, \ldots, e_n) = \varepsilon(\sigma)$. Therefore $f(v_1, \ldots, v_n) = \sum_\sigma \varepsilon(\sigma) \prod_j a_{\sigma(j),j} = \det(X)$: f is forced to be the determinant. ■ *(and that's on periodt)*

So the three rules of Section 1 are not a vibe: they pin down the determinant completely. Any function obeying them IS the determinant.

2.7 Transpose and product

THEOREM / PROPOSITION // Transpose

For every $A \in \mathcal{M}_n(\mathbb{K})$: $\det(A^\top) = \det(A)$. Consequently every column rule is also a row rule.

RECEIPTS (proof) // reindex by σ^{-1}

$\det(A^\top) = \sum_\sigma \varepsilon(\sigma) \prod_{j=1}^{n} (A^\top)_{\sigma(j),j} = \sum_\sigma \varepsilon(\sigma) \prod_{j=1}^{n} a_{j,\sigma(j)}$. Put $\tau = \sigma^{-1}$ (a bijection of $\mathfrak{S}_n$) and use $\varepsilon(\tau) = \varepsilon(\sigma)$. In the product, set $k = \sigma(j)$, i.e. $j = \tau(k)$: $\prod_j a_{j,\sigma(j)} = \prod_k a_{\tau(k),k}$. So $\det(A^\top) = \sum_\tau \varepsilon(\tau) \prod_k a_{\tau(k),k} = \det A$. ■ *(and that's on periodt)*

THEOREM / PROPOSITION // Product

For $A, B \in \mathcal{M}_n(\mathbb{K})$: $\det(AB) = \det(A) \det(B)$.

RECEIPTS (proof) // use the characterisation

Fix A and set $f(C_1, \ldots, C_n) = \det(AC_1, \ldots, AC_n)$. Since $C \mapsto AC$ is linear and det is multilinear, f is multilinear; if $C_i = C_j$ then $AC_i = AC_j$ and $f = 0$: alternating. By the uniqueness argument (which works for any multilinear alternating f, giving $f = f(e_1, \ldots, e_n) \cdot \det$), $f(C_1, \ldots, C_n) = c \det(C_1, \ldots, C_n)$ with $c = f(e_1, \ldots, e_n) = \det(Ae_1, \ldots, Ae_n) = \det A$. Take $C_1, \ldots, C_n =$ the columns of B: the columns of AB are $AC_1, \ldots, AC_n$, so $\det(AB) = \det(A) \det(B)$. ■ *(and that's on periodt)*

2.8 Boss level: Laplace expansion from Leibniz

Path B uses *cofactor expansion*: $\det A = \sum_{i=1}^{n} (-1)^{i+j} a_{ij} \det M_{ij}$, where M_{ij} is A without row i and column j. Here is why it's true.

RECEIPTS (proof) // Laplace expansion along column j (optional)

Group the Leibniz sum according to the row $i = \sigma(j)$ chosen in column j:

$$\det A = \sum_{i=1}^{n} a_{ij} \sum_{\sigma:\ \sigma(j)=i} \varepsilon(\sigma) \prod_{k \neq j} a_{\sigma(k),k}.$$

Remove position j from the inputs and value i from the outputs, and relabel both remaining sets as $1, \ldots, n-1$ in increasing order. Each σ with $\sigma(j) = i$ becomes exactly one $\pi \in \mathfrak{S}_{n-1}$, and $\prod_{k \neq j} a_{\sigma(k),k}$ is precisely the Leibniz term of π for the minor M_{ij}.
Signs: inversions of σ not involving position j are exactly those of π. Those involving j number $a + b$ with $a = \#\{k < j : \sigma(k) > i\}$ and $b = \#\{k > j : \sigma(k) < i\}$. With $c = \#\{k < j : \sigma(k) < i\}$ we have $a + c = j - 1$ and $b + c = i - 1$, so $a + b = i + j - 2 - 2c$ has the parity of $i + j$. Hence $\varepsilon(\sigma) = (-1)^{i+j}\varepsilon(\pi)$, the inner sum equals $(-1)^{i+j} \det M_{ij}$, and the formula follows. Expansion along a *row* follows by applying this to $A^\top$. ■ *(and that's on periodt)*

TL;DR // Path A recap

1. $\mathfrak{S}_n$ = permutations of $\{1, \ldots, n\}$, $n!$ of them; each is a product of transpositions.
2. $\varepsilon(\sigma) = (-1)^{\#\text{inversions}} = (-1)^{\#\text{transpositions}}$, multiplicative, $\varepsilon(\tau) = -1$.
3. $\det A = \sum_{\sigma \in \mathfrak{S}_n} \varepsilon(\sigma) \prod_j a_{\sigma(j),j}$: one entry per row and column, signed.
4. From Leibniz: multilinear, alternating, $\det I_n = 1$, and it's the *only* such function. Then $\det A^\top = \det A$, $\det AB = \det A \det B$, and Laplace expansion.

3 Path B – Speedrun: how to compute any determinant

PATH B: SPEEDRUN

TL;DR // Path B in one breath

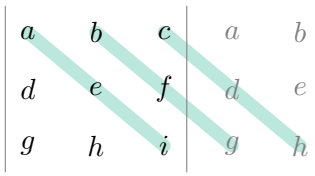
1. 2×2: $ad-bc$. 3×3: Sarrus. Bigger: **expand along a row or column** using **cofactors** (minor $\times$ sign from the $+-+-$ grid).
2. Before expanding, **create zeros** with row/column operations (adding a multiple of a row to another changes nothing).
3. Triangular matrix: multiply the diagonal. Done.
4. $\det A = 0 \iff$ the rows (or columns) are **linearly dependent**. A zero row, two equal or proportional rows are quick ways to *see* it, but not the only ones.

3.1 Small sizes: 1×1, 2×2, 3×3

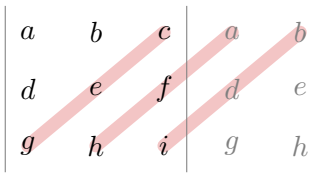
DEFINITION // Small determinants

$$\det(a) = a, \qquad \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad-bc, \qquad \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei+bfg+cdh-ceg-bdi-afh.$$

Rule of Sarrus (how to remember the 3×3 formula): copy the first two columns to the right. The three "down-right" diagonals are counted with $+$, the three "up-right" diagonals with $-$.



$+\,aei \;+\, bfg \;+\, cdh$



$-\,ceg \;-\, afh \;-\, bdi$

Copy columns 1 and 2 (grey) to the right of the matrix. Left: the three diagonals going down-right give the $+$ terms. Right: the three diagonals going up-right give the $-$ terms.

COMMON L (don't do this) // Sarrus is ONLY for 3×3

There is no "Sarrus for 4×4": a 4×4 determinant has 24 terms, the diagonal trick only gives 8. For $n\geq 4$, use cofactor expansion and row operations.

3.2 Minors and cofactors

DEFINITION // Minor matrix, minor, cofactor of a coefficient

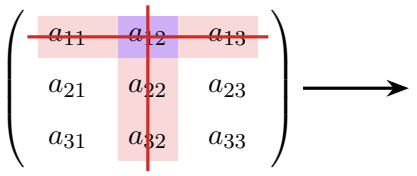
Let $A=(a_{ij})\in\mathcal{M}_n(\mathbb{R})$ and fix a position (i,j).

- The **minor matrix** of the coefficient a_{ij}, written M_{ij}, is the $(n-1)\times(n-1)$ matrix obtained by **deleting row i and column j** from A.
- The **minor** of a_{ij} is the number $\det(M_{ij})$. (Some books call M_{ij} itself "the minor"; always

say which one you mean.)

- The **cofactor** of a_{ij} is $C_{ij} = (-1)^{i+j} \det(M_{ij})$: the minor with a sign.

delete row 1 and column 2



$$M_{12} = \begin{pmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{pmatrix} \qquad C_{12} = (-1)^{1+2} \det M_{12} = -(a_{21}a_{33} - a_{23}a_{31})$$

EXAMPLE // Minors and cofactors of row 1

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 1 & 2 & 0 \end{pmatrix}.$$

1. $(1,1)$: delete row 1, column 1: $M_{11} = \begin{pmatrix} -1 & 4 \\ 2 & 0 \end{pmatrix}$, $\det M_{11} = (-1)(0) - (4)(2) = -8$, sign $(-1)^2 = +$, so $C_{11} = -8$.
2. $(1,2)$: $M_{12} = \begin{pmatrix} 0 & 4 \\ 1 & 0 \end{pmatrix}$, $\det M_{12} = 0 - 4 = -4$, sign $(-1)^3 = -$, so $C_{12} = +4$.
3. $(1,3)$: $M_{13} = \begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix}$, $\det M_{13} = 0 - (-1) = 1$, sign $(-1)^4 = +$, so $C_{13} = 1$.

3.3 The $+ - +-$ sign grid

The sign $(-1)^{i+j}$ only depends on the *position*. It makes a chessboard that always starts with $+$ in the top-left corner:

+	−	+	−	+
−	+	−	+	−
+	−	+	−	+
−	+	−	+	−
+	−	+	−	+

sign of position $(i,j) = (-1)^{i+j}$:

- $+$ when $i+j$ is even, $-$ when odd;
- the main diagonal is always $+$;
- neighbours (left/right/up/down) alternate.

Example: position $(2,3)$: $2+3=5$ odd $\Rightarrow -$.

3.4 Cofactor (Laplace) expansion

THEOREM / PROPOSITION // Laplace expansion

For $A \in \mathcal{M}_n(\mathbb{R})$, for **any** fixed row i and **any** fixed column j:

$$\text{along row } i: \qquad \det(A) = \sum_{j=1}^{n} a_{ij}\, C_{ij} = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} \det M_{ij},$$

$$\text{along column } j: \qquad \det(A) = \sum_{i=1}^{n} a_{ij}\, C_{ij} = \sum_{i=1}^{n} (-1)^{i+j} a_{ij} \det M_{ij}.$$

Every row and every column gives the same answer. (Proof: Path A, "boss level".)

EXAM CHEAT CODE // "Compute $\det A$ by cofactor expansion"

1. **Pick the row or column with the most zeros** (every zero kills a whole term).
2. For each non-zero entry a_{ij} of that line:
   (a) delete row i and column j to get M_{ij};
   (b) compute $\det M_{ij}$ (a smaller determinant: recurse);
   (c) attach the sign $(-1)^{i+j}$ from the grid.
3. Add up $a_{ij} \cdot (\pm \det M_{ij})$.
4. Sanity check if you have time: expand along a different line.

EXAMPLE // Same 3×3, two expansions

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 1 & 2 & 0 \end{pmatrix}.$$

Along row 1 (cofactors computed above): $\det A = 2 \cdot C_{11} + 1 \cdot C_{12} + 3 \cdot C_{13} = 2(-8) + 1(4) + 3(1) = -9$.

Along column 1 (it has a zero, so only 2 terms):

1. $a_{11} = 2$, sign $+$, $\det M_{11} = -8$: contributes -16.
2. $a_{21} = 0$: contributes 0, no need to compute anything.
3. $a_{31} = 1$, sign $(-1)^{3+1} = +$, $M_{31} = \begin{pmatrix} 1 & 3 \\ -1 & 4 \end{pmatrix}$, $\det M_{31} = 4 + 3 = 7$: contributes 7.

$\det A = -16 + 0 + 7 = -9$. ✓ Same as Sarrus in Section 1.

EXAMPLE // A 4×4 that's secretly easy

$$A = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 3 & 0 & 4 & 5 \\ 0 & 0 & 6 & 0 \\ 2 & 7 & 1 & 3 \end{pmatrix}.$$

1. Column 2 has a single non-zero entry: $a_{42} = 7$, at position $(4, 2)$, sign $(-1)^6 = +$. So $\det A = 7 \det M_{42}$ with $M_{42} = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 4 & 5 \\ 0 & 6 & 0 \end{pmatrix}$.
2. In M_{42}, row 3 has a single non-zero entry: 6 at position $(3, 2)$, sign $(-1)^5 = -$. So $\det M_{42} = -6 \begin{vmatrix} 1 & 0 \\ 3 & 5 \end{vmatrix} = -6 \cdot 5 = -30$.
3. $\det A = 7 \cdot (-30) = -210$.

Two easy steps instead of 24 Leibniz terms.

3.5 Row and column operations: the real power move

THEOREM / PROPOSITION // What each operation does to det

Operation (rows or columns)	**Effect on** det	**Why**
swap two rows: $R_i \leftrightarrow R_j$	multiplied by -1	alternating (R2)
scale a row: $R_i \leftarrow \lambda R_i$	multiplied by λ	linear in each row (R3)
add a multiple of another row: $R_i \leftarrow R_i + \lambda R_j$	**unchanged**	(C3)
transpose	unchanged	$\det A^\top = \det A$

Equivalently, "factor out" a common factor λ from one row or column: $\det(\dots, \lambda C, \dots) = \lambda \det(\dots, C, \dots)$.

COMMON L (don't do this) // Row-operation traps

- $R_i \leftarrow \lambda R_i + \mu R_j$ multiplies the determinant by λ (it's "scale by λ" and "add μR_j" in one go). Prefer the clean form $R_i \leftarrow R_i + \mu R_j$.
- In Lecture 2, scaling a row didn't change the *solutions* of a system. Here it **does** change the determinant: keep track of every factor you pull out or multiply in.
- Do operations one at a time, each using the *current* rows (or state clearly that you use the original ones).
- You may mix row and column operations freely for a determinant (unlike when solving a system).

THEOREM / PROPOSITION // Triangular and diagonal matrices

If T is upper or lower triangular (in particular diagonal), $\det T$ is the **product of its diagonal entries**: $\det T = t_{11} t_{22} \cdots t_{nn}$.

RECEIPTS (proof) // induction, expanding along the first column

For $n = 1$ it's clear. If T is upper triangular, its first column is $(t_{11}, 0, \dots, 0)^\top$; expanding along it, $\det T = t_{11} \det M_{11}$, and M_{11} is upper triangular with diagonal $t_{22}, \dots, t_{nn}$: induction. Lower triangular: transpose. ■ *(and that's on periodt)*

EXAM CHEAT CODE // "Compute $\det A$ by elimination" (the fastest method for big matrices)

1. Use $R_i \leftarrow R_i + \lambda R_j$ (free!) and row swaps (each costs a factor -1) to make the matrix **upper triangular**, exactly like Gaussian elimination in Lecture 2.
2. $\det A = (-1)^{\#\text{swaps}} \times$ (product of the diagonal).
3. If you also scaled rows or pulled out factors, multiply those factors back in.

EXAMPLE // Elimination on the same 3×3

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 1 & 2 & 0 \end{pmatrix}.$$

1. $R_3 \leftarrow R_3 - \frac{1}{2}R_1$: row 3 becomes $(0,\ \frac{3}{2},\ -\frac{3}{2})$.
2. $R_3 \leftarrow R_3 + \frac{3}{2}R_2$: row 3 becomes $(0,\ 0,\ -\frac{3}{2}+6) = (0, 0, \frac{9}{2})$.
3. Triangular with diagonal $2, -1, \frac{9}{2}$, no swaps: $\det A = 2 \cdot (-1) \cdot \frac{9}{2} = -9$. ✓

EXAMPLE // A 4×4 with a trick: add all the columns

$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix}$. Every row sums to 10: that's a hint.

1. $C_1 \leftarrow C_1 + C_2 + C_3 + C_4$ (unchanged det): the first column becomes $(10, 10, 10, 10)$. Factor out 10:
$$\det A = 10 \begin{vmatrix} 1 & 2 & 3 & 4 \\ 1 & 3 & 4 & 1 \\ 1 & 4 & 1 & 2 \\ 1 & 1 & 2 & 3 \end{vmatrix}.$$
2. $R_2 \leftarrow R_2 - R_1$, $R_3 \leftarrow R_3 - R_1$, $R_4 \leftarrow R_4 - R_1$ (unchanged):
$$\det A = 10 \begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & -3 \\ 0 & 2 & -2 & -2 \\ 0 & -1 & -1 & -1 \end{vmatrix}.$$
3. Expand along column 1 (only the 1 on top survives, sign $+$):
$$\det A = 10 \begin{vmatrix} 1 & 1 & -3 \\ 2 & -2 & -2 \\ -1 & -1 & -1 \end{vmatrix}.$$
4. Factor 2 out of row 2 and -1 out of row 3:
$$\det A = 10 \cdot 2 \cdot (-1) \begin{vmatrix} 1 & 1 & -3 \\ 1 & -1 & -1 \\ 1 & 1 & 1 \end{vmatrix}.$$
5. $R_2 \leftarrow R_2 - R_1$, $R_3 \leftarrow R_3 - R_1$: $\begin{vmatrix} 1 & 1 & -3 \\ 0 & -2 & 2 \\ 0 & 0 & 4 \end{vmatrix} = 1 \cdot (-2) \cdot 4 = -8$ (triangular).
6. $\det A = 10 \cdot 2 \cdot (-1) \cdot (-8) = 160$.

3.6 When is a determinant zero? (read this carefully)

THEOREM / PROPOSITION // THE criterion

For $A \in \mathcal{M}_n(\mathbb{R})$, the following are **equivalent**:

1. $\det A = 0$;
2. the columns of A are **linearly dependent** (one column is a linear combination of the others);

3. the rows of A are linearly dependent;
4. A is not invertible; $\operatorname{rank} A < n$; $\ker A \neq \{0\}$ (some $x \neq 0$ has $Ax = 0$).

(Proof in Section 4. Rows and columns are interchangeable because $\det A^\top = \det A$.)

THEOREM / PROPOSITION // Quick ways to SPOT a zero determinant

Each of these is **enough** to conclude $\det A = 0$:

(a) a row (or column) is **all zeros**;

(b) two rows (or two columns) are **equal**;

(c) two rows (or two columns) are **proportional** ($R_i = \lambda R_j$);

(d) a row (or column) is a **combination** of other rows, e.g. $R_3 = R_1 + R_2$.

All four are special cases of "the rows (columns) are linearly dependent".

COMMON L (don't do this) // "if", NOT "if and only if"

It is **false** that "$\det A = 0$ if and only if a row is zero" or "if and only if two rows are proportional". Those situations *force* $\det A = 0$, but $\det A$ can be 0 without any of them:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} : \quad \text{no zero row, no two proportional rows, yet } \det A = 0,$$

because $R_1 - 2R_2 + R_3 = (1-8+7,\ 2-10+8,\ 3-12+9) = (0,0,0)$: the rows are dependent in a "hidden" way. The only correct "if and only if" is: $\det A = 0 \iff$ **rows (or columns) linearly dependent**. Equivalently: $\det A \neq 0 \iff$ rows independent.

EXAM CHEAT CODE // "Show that $\det A = 0$" / "For which x is $\det A(x) = 0$?"

1. **Look first:** zero line? equal or proportional lines? one line = sum/difference of others?
2. **Prove it cleanly:** exhibit a non-zero x with $Ax = 0$ (here $x = (1,-2,1)$: $Ax = (1-4+3,\ 4-10+6,\ 7-16+9) = 0$) or write the explicit dependence between rows.
3. **With a parameter x:** find the values that make two rows equal or proportional; each such value x_0 gives a factor $(x - x_0)$ of the polynomial $\det A(x)$. Then compute and *check* the factors are there (this is exactly how Exercise 3 is solved).

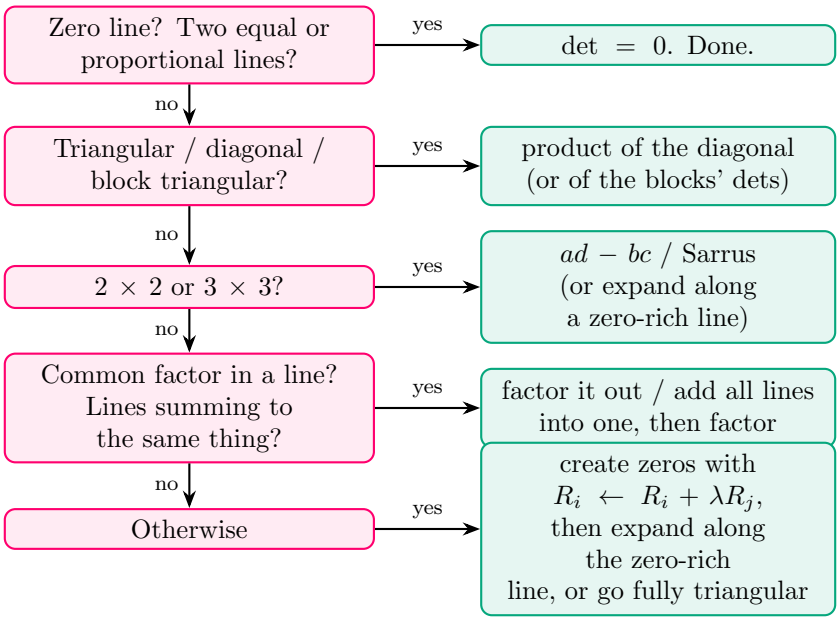
3.7 Instant determinants (memorise these)

Matrix	**Determinant**
I_n	1
diagonal $\operatorname{diag}(\lambda_1, \dots, \lambda_n)$	$\lambda_1 \lambda_2 \cdots \lambda_n$
triangular (upper or lower)	product of the diagonal
block triangular $\begin{pmatrix} A & B \\ 0 & D \end{pmatrix}$ (A, D square)	$\det A \cdot \det D$ (Exercise 7)
$A^\top$	$\det A$
AB	$\det A \cdot \det B$
A^{-1}	$1/\det A$
λA $(n \times n)$	$\lambda^n \det A$
A^k	$(\det A)^k$
orthogonal U $(U^\top U = I)$	± 1
permutation matrix of σ	$\varepsilon(\sigma)$ (Path A)
Vandermonde $\left(x_j^{i-1}\right)_{i,j}$	$\prod_{1 \le i < j \le n} (x_j - x_i)$ (Exercise 8)

EXAMPLE // Instant answers

1. $\begin{vmatrix} 3 & 7 & -2 \\ 0 & -1 & 5 \\ 0 & 0 & 4 \end{vmatrix} = 3 \cdot (-1) \cdot 4 = -12$ (upper triangular).
2. If $\det A = 2$ for $A \in \mathcal{M}_3(\mathbb{R})$: $\det(3A) = 3^3 \cdot 2 = 54$, $\det(A^{-1}) = \frac{1}{2}$, $\det(A^\top A) = 2 \cdot 2 = 4$, $\det(A^5) = 32$.
3. $\begin{vmatrix} 1 & 2 & 0 & 0 \\ 3 & 4 & 0 & 0 \\ 5 & 6 & 2 & 0 \\ 7 & 8 & 1 & 3 \end{vmatrix}$ is block lower triangular: $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} \cdot \begin{vmatrix} 2 & 0 \\ 1 & 3 \end{vmatrix} = (4-6)(6-0) = -12$.

3.8 Strategy: how to attack any determinant



3.9 Toward the characteristic polynomial (why Path B matters next week)

In Section 6 we'll see that λ is an *eigenvalue* of A (some $v \neq 0$ has $Av = \lambda v$) exactly when $\det(A - \lambda I_n) = 0$. So finding eigenvalues = computing a determinant **with a variable λ inside**, then finding its roots. All the tricks above are what make this doable by hand.

DEFINITION // Characteristic polynomial

For $A \in \mathcal{M}_n(\mathbb{R})$, the **characteristic polynomial** is $\chi_A(\lambda) = \det(A - \lambda I_n)$: subtract λ from every diagonal entry, take the determinant. It is a polynomial of degree n in λ. (Some books use $\det(\lambda I_n - A) = (-1)^n \chi_A(\lambda)$: same roots.)

THEOREM / PROPOSITION // The 2×2 shortcut

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$: $\chi_A(\lambda) = \begin{vmatrix} a-\lambda & b \\ c & d-\lambda \end{vmatrix} = \lambda^2 - \operatorname{tr}(A)\,\lambda + \det(A)$, where $\operatorname{tr}(A) = a + d$ is the **trace** (sum of the diagonal).

RECEIPTS (proof) //

$(a-\lambda)(d-\lambda) - bc = \lambda^2 - (a+d)\lambda + (ad - bc)$. ■ *(and that's on periodt)*

EXAMPLE // 2×2 characteristic polynomial

$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$: $\operatorname{tr} A = 4$, $\det A = 4 - 1 = 3$, so $\chi_A(\lambda) = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3)$. Eigenvalues: 1 and 3. (Same matrix as Lecture 2, where the basis $(1, 1), (1, -1)$ made it $\operatorname{diag}(3, 1)$. Not a coincidence.)

EXAMPLE // 3×3: factor, don't expand blindly

$A = \begin{pmatrix} 4 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 4 \end{pmatrix}$, so $A - \lambda I_3 = \begin{pmatrix} 4-\lambda & 1 & 1 \\ 1 & 4-\lambda & 1 \\ 1 & 1 & 4-\lambda \end{pmatrix}$.

1. Every row sums to $6 - \lambda$. Do $C_1 \leftarrow C_1 + C_2 + C_3$ (unchanged det): the first column becomes $(6-\lambda)(1,1,1)^\top$. Factor it out:

$$\chi_A(\lambda) = (6-\lambda) \begin{vmatrix} 1 & 1 & 1 \\ 1 & 4-\lambda & 1 \\ 1 & 1 & 4-\lambda \end{vmatrix}.$$

2. $R_2 \leftarrow R_2 - R_1$, $R_3 \leftarrow R_3 - R_1$ (unchanged):

$$\chi_A(\lambda) = (6-\lambda) \begin{vmatrix} 1 & 1 & 1 \\ 0 & 3-\lambda & 0 \\ 0 & 0 & 3-\lambda \end{vmatrix}.$$

3. Triangular: $\chi_A(\lambda) = (6-\lambda)(3-\lambda)^2$.

4. Eigenvalues: 6 (simple) and 3 (double). Already factored, no cubic equation to solve.

Sanity checks: $\chi_A(0) = \det A$, here $6 \cdot 9 = 54$; and over $\mathbb{C}$ the eigenvalues (with multiplicity) sum to the trace and multiply to the determinant: $6+3+3 = 12 = 4+4+4$ ✓, $6 \cdot 3 \cdot 3 = 54 = \det A$ ✓.

COMMON L (don't do this) // Characteristic-polynomial traps

- Subtract λ **only on the diagonal**, not everywhere.
- Don't expand everything into $-\lambda^3 + \dots$ and then try to factor a cubic. Keep factors as you go (sum-of-lines trick, triangularisation, expansion along a line with zeros).
- For a **triangular** matrix, $\chi_A(\lambda) = \prod_i (a_{ii} - \lambda)$: the eigenvalues are the diagonal entries.

3.10 Bonus: cofactors give the inverse

THEOREM / PROPOSITION // Comatrix formula

Let $\mathrm{Com}(A) = (C_{ij})$ be the matrix of cofactors. Then $A\,\mathrm{Com}(A)^\top = \det(A)\, I_n$. So if $\det A \neq 0$: $A^{-1} = \dfrac{1}{\det A}\,\mathrm{Com}(A)^\top$.

RECEIPTS (proof) //

Entry (i,k) of $A\,\mathrm{Com}(A)^\top$ is $\sum_j a_{ij} C_{kj}$. If $k = i$, this is the expansion of $\det A$ along row i. If $k \neq i$, it is the expansion along row k of the matrix obtained from A by replacing row k with row i; that matrix has two equal rows, so it is 0. ■ *(and that's on periodt)*

For 2×2: $\mathrm{Com}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}$, so $A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. That's the Lecture 2 formula, now with receipts. For $n \geq 3$, Gauss–Jordan is usually faster.

TL;DR // Path B recap

1. Minor matrix M_{ij}: delete row i, column j. Cofactor $C_{ij} = (-1)^{i+j} \det M_{ij}$, sign from the $+-+-$ grid.
2. $\det A = \sum_j a_{ij} C_{ij}$ along any row, or $\sum_i a_{ij} C_{ij}$ along any column. Choose the line with most zeros.
3. $R_i \leftarrow R_i + \lambda R_j$: free. Swap: $\times(-1)$. Scale a line by λ: $\times\lambda$.
4. Triangular: product of the diagonal. $\det(\lambda A) = \lambda^n \det A$. $\det(AB) = \det A \det B$.
5. $\det A = 0 \iff$ lines dependent. Zero / equal / proportional lines are sufficient signs, not the definition.
6. $\chi_A(\lambda) = \det(A - \lambda I)$: keep it factored. 2×2: $\lambda^2 - \operatorname{tr}(A)\lambda + \det A$.

4 The properties everyone needs

EVERYONE *Both paths meet here. Path A readers have seen the proofs of the starred ($\star$) rules; Path B readers can take them as given.*

THEOREM / PROPOSITION // The determinant rulebook

For $A, B \in \mathcal{M}_n(\mathbb{K})$ and $\lambda \in \mathbb{K}$:

1. $\det(I_n) = 1$.
2. $\det(A^\top) = \det(A)$ ($\star$): every rule about columns is also true for rows.
3. $\det(AB) = \det(A)\det(B)$ ($\star$): the determinant of a product is the product of the determinants.
4. $\det(\lambda A) = \lambda^n \det(A)$.
5. det is linear in one row (or column) when the others are fixed ($\star$).
6. Swapping two rows (or columns) multiplies det by -1 ($\star$).
7. Adding a multiple of one row (column) to another leaves det unchanged.
8. A invertible $\iff \det A \neq 0$, and then $\det(A^{-1}) = \dfrac{1}{\det A}$.

4.1 Determinant and invertibility

THEOREM / PROPOSITION // Invertibility criterion

A is invertible $\iff \det(A) \neq 0$. Equivalently: A is singular $\iff \det(A) = 0$.

RECEIPTS (proof) // two directions

($\Rightarrow$) If A is invertible, $1 = \det(I_n) = \det(AA^{-1}) = \det(A)\det(A^{-1})$, so $\det A \neq 0$ (and $\det A^{-1} = 1/\det A$ for free).
($\Leftarrow$, by contraposition) If A is not invertible, its columns are linearly dependent (Lecture 2): some column is a combination of the others, $C_j = \sum_{k \neq j} \alpha_k C_k$. By linearity in column j,

$\det A = \sum_{k\neq j} \alpha_k \det(C_1, \dots, \underbrace{C_k}_{\text{position } j}, \dots, C_n)$, and each determinant on the right contains C_k twice, hence is 0. So $\det A = 0$. ■ *(and that's on periodt)*

This also proves THE criterion of Path B: $\det A = 0 \iff$ *not invertible* $\iff$ *columns dependent* $\iff$ *rows dependent (apply it to* $A^\top$*)* $\iff \operatorname{rank} A < n \iff \ker A \neq \{0\}$*, by the big theorem of Lecture 2.*

ML SIDE QUEST // Determinant and information loss

For a feature transformation $z = Ax$ with $A \in \mathbb{R}^{d\times d}$: if $\det A \neq 0$, A is invertible and $x = A^{-1}z$ can be recovered; if $\det A = 0$, A collapses at least one direction and some information is lost. The determinant detects whether a square linear representation **preserves or destroys information**.

4.2 Determinant of a family of vectors, change of basis

DEFINITION // Determinant of a family in a basis

Let E have dimension n and basis $\mathcal{B}$. For $v_1, \dots, v_n \in E$, $\det_{\mathcal{B}}(v_1, \dots, v_n) = \det(X)$ where $X = \big[\, [v_1]_{\mathcal{B}} \mid \cdots \mid [v_n]_{\mathcal{B}} \,\big]$ is the matrix of their coordinates.

THEOREM / PROPOSITION // Basis criterion and change of basis

1. $(v_1, \dots, v_n)$ is a basis of $E \iff \det_{\mathcal{B}}(v_1, \dots, v_n) \neq 0$.
2. If $\mathcal{B}'$ is another basis and P is the change-of-basis matrix (columns $[e'_j]_{\mathcal{B}}$, so that $[x]_{\mathcal{B}} = P[x]_{\mathcal{B}'}$), then
$$\det_{\mathcal{B}}(v_1, \dots, v_n) = \det(P)\, \det_{\mathcal{B}'}(v_1, \dots, v_n),$$
equivalently $\det_{\mathcal{B}'}(v_1, \dots, v_n) = \dfrac{1}{\det P} \det_{\mathcal{B}}(v_1, \dots, v_n)$.

RECEIPTS (proof) //

1. The family is a basis iff the coordinate matrix X has independent columns iff X is invertible iff $\det X \neq 0$. *2.* Let X' be the coordinate matrix in $\mathcal{B}'$. Column by column, $[v_j]_{\mathcal{B}} = P[v_j]_{\mathcal{B}'}$, so $X = PX'$ and $\det X = \det P \det X'$. P is invertible, so $\det P \neq 0$ and we can divide. ■ *(and that's on periodt)*

EXAMPLE // Is it a basis?

$v_1 = (1,1,0)$, $v_2 = (0,1,1)$, $v_3 = (1,0,1)$ in $\mathbb{R}^3$ (the basis $\mathcal{B}'$ of Lecture 2's exercise). $\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix}$: by Sarrus, $1 + 0 + 1 - 0 - 0 - 0 = 2 \neq 0$. So yes, it's a basis.

4.3 Determinant of an endomorphism

THEOREM / PROPOSITION // Similar matrices have the same determinant

If $B = P^{-1}AP$ with P invertible, then $\det B = \det A$.

RECEIPTS (proof) //

$\det B = \det(P^{-1})\det(A)\det(P) = \frac{1}{\det P}\det(A)\det(P) = \det A$. (Numbers commute, even if matrices don't.) ■ *(and that's on periodt)*

DEFINITION // Determinant of an endomorphism

Let $f : E \to E$ be linear, E finite-dimensional. $\det(f)$ is the determinant of the matrix of f in **any** basis of E. By the previous theorem and Lecture 2's basis-change formula $[f]_{\mathcal{B}'} = P^{-1}[f]_{\mathcal{B}}P$, this does not depend on the chosen basis: it's a property of the map itself, not of a coordinate system.

THEOREM / PROPOSITION // Properties for endomorphisms

For $f, g : E \to E$ linear:

1. f is bijective $\iff \det(f) \neq 0$;
2. $\det(f \circ g) = \det(f)\det(g)$;
3. for every basis $\mathcal{B} = (v_1, \dots, v_n)$: $\det_{\mathcal{B}}(f(v_1), \dots, f(v_n)) = \det(f)\det_{\mathcal{B}}(v_1, \dots, v_n) = \det(f)$;
4. if f is bijective, $\det(f^{-1}) = \dfrac{1}{\det f}$.

RECEIPTS (proof) // translate everything into matrices

Fix $\mathcal{B}$, $A = [f]_{\mathcal{B}}$, $B = [g]_{\mathcal{B}}$. (1) f bijective $\iff A$ invertible $\iff \det A \neq 0$. (2) $[f \circ g]_{\mathcal{B}} = AB$ (Lecture 2), so $\det(f \circ g) = \det A \det B$. (3) The coordinate matrix of $(f(v_1), \dots, f(v_n))$ in $\mathcal{B}$ is exactly A (columns = images of the basis vectors), and $\det_{\mathcal{B}}(v_1, \dots, v_n) = \det I_n = 1$. (4) $f \circ f^{-1} = \mathrm{Id}_E$ gives $\det(f)\det(f^{-1}) = 1$. ■ *(and that's on periodt)*

EXAMPLE // Determinant of a map that's not a matrix: the derivative

$D : \mathbb{R}_2[X] \to \mathbb{R}_2[X]$, $P \mapsto P'$ had matrix $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$ in the basis $(1, X, X^2)$ (Lecture 2). Upper triangular with diagonal $0, 0, 0$: $\det D = 0$. Consistent: D is not injective (constants have derivative 0). Exercise 4 does the same for $M \mapsto M^\top$.

5 Determinants of some particular matrices

EVERYONE

THEOREM / PROPOSITION // Diagonal and triangular matrices

$\det(\mathrm{diag}(\lambda_1, \ldots, \lambda_n)) = \prod_{i=1}^{n} \lambda_i$, and the same for any triangular matrix (product of the diagonal). **Corollary:** a diagonal (or triangular) matrix is invertible iff all its diagonal entries are non-zero.

RECEIPTS (proof) // two ways

Path A way: in Leibniz, $\prod_j d_{\sigma(j),j}$ contains an off-diagonal entry ($= 0$) unless $\sigma(j) = j$ for all j, so only $\sigma = \mathrm{Id}$ survives. *Path B way:* expand along the first column and induct (Section 3). Corollary: a finite product is non-zero iff every factor is. ■ *(and that's on periodt)*

5.1 Orthogonal matrices: determinant ± 1

THEOREM / PROPOSITION // Determinant of an orthogonal matrix

If $U \in \mathbb{R}^{n \times n}$ is orthogonal ($U^\top U = I_n$, so $U^{-1} = U^\top$), then $\det U \in \{-1, +1\}$.

RECEIPTS (proof) //

$\det(U^\top)\det(U) = \det(I_n) = 1$ and $\det U^\top = \det U$, so $(\det U)^2 = 1$. Over $\mathbb{R}$: $\det U = \pm 1$. ■ *(and that's on periodt)*

Orthogonal matrices preserve volumes ($|\det| = 1$). $\det U = +1$: orientation preserved (rotations). $\det U = -1$: orientation flipped (reflections).

EXAMPLE // Rotations, reflections, permutations

1. **Rotation** by θ: $R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, $\det R(\theta) = \cos^2\theta + \sin^2\theta = 1$. Also $R(0) = I_2$, $R(\theta_1)R(\theta_2) = R(\theta_1 + \theta_2)$, $R(\theta)^{-1} = R(\theta)^\top = R(-\theta)$.
2. **Reflection** across the x-axis: $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, determinant -1.
3. **Permutation matrix** (swap coordinates): $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, determinant -1 (one transposition).
4. **Numbers** ($\theta = \pi/6$): $R(\frac{\pi}{6}) = \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$, $R(\frac{\pi}{6})\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \sqrt{3}/2 \\ 1/2 \end{pmatrix}$, of norm $\sqrt{\frac{3}{4} + \frac{1}{4}} = 1$: the length is preserved.

THEOREM / PROPOSITION // Why "orthogonal"

If U is orthogonal: its columns (and rows) form an **orthonormal basis** of $\mathbb{R}^n$, and U preserves dot products and norms: $(Ux) \cdot (Uy) = x \cdot y$, $\|Ux\|_2 = \|x\|_2$. (Proved in Lecture 2; norms are the topic of Lecture 5.)

ML SIDE QUEST // Rotations in transformers (RoPE)

Rotary position embeddings (RoPE, used in many large language models) encode the position of a token by **rotating** pairs of coordinates of its query/key vectors by an angle proportional to the position, with matrices $R(\theta)$ exactly like above. Because rotations are orthogonal ($\det = 1$), they don't change lengths, and $R(\theta_1)^\top R(\theta_2) = R(\theta_2 - \theta_1)$ makes the attention score depend

only on the *relative* position.

6 Eigenvalues, eigenvectors and the start of diagonalisation

EVERYONE

TL;DR // Chapter 6 in one breath

An **eigenvector** of A is a non-zero vector that A only *stretches*: $Av = \lambda v$. The stretch factor λ is the **eigenvalue**. The eigenvalues are the roots of $\det(A - \lambda I)$, which is why we needed determinants. If you can find n independent eigenvectors, then in that basis A is just a diagonal matrix: $A = VDV^{-1}$.

6.1 Definitions

DEFINITION // Eigenvalue, eigenvector

Let $A \in \mathbb{R}^{n\times n}$. A **non-zero** vector $v \in \mathbb{R}^n$ is an **eigenvector** of A if there exists $\lambda \in \mathbb{R}$ with

$$Av = \lambda v.$$

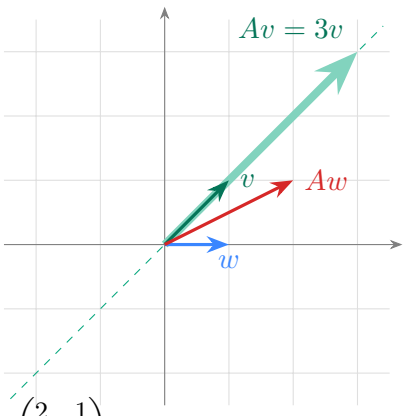
The scalar λ is the **eigenvalue** associated with v.

COMMON L (don't do this) // $v \neq 0$ is part of the definition

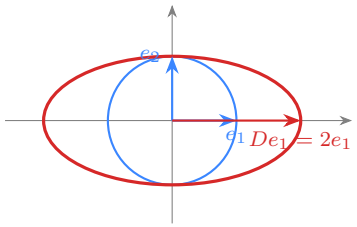
$A0 = \lambda 0$ for every λ, so allowing $v = 0$ would make every number an eigenvalue. Eigen*vectors* are non-zero. (Eigen*values*, on the other hand, can be 0: that happens exactly when A is not invertible.)

EXAMPLE // The easy cases

1. I_n: every x satisfies $I_n x = 1 \cdot x$, so **every non-zero vector** is an eigenvector, with eigenvalue 1.
2. $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$: $De_i = \lambda_i e_i$, so the canonical vector e_i is an eigenvector with eigenvalue λ_i.



$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$: $v = (1,1)$ stays on its line, $w = (1,0)$ changes direction



$D = \operatorname{diag}(2,1)$: unit circle $\to$ ellipse $\frac{y_1^2}{4} + y_2^2 = 1$; axes = eigenvectors

EXAMPLE // Where does the ellipse come from?

Take x on the unit circle ($x_1^2 + x_2^2 = 1$) and $y = Dx = (2x_1, x_2)$.

1. Solve for x: $x_1 = y_1/2$, $x_2 = y_2$.
2. Substitute into $x_1^2 + x_2^2 = 1$: $\frac{y_1^2}{4} + y_2^2 = 1$, the equation of an ellipse with half-axes 2 (along e_1) and 1 (along e_2).

DEFINITION // Eigenspace, eigenbasis, spectrum

- The **eigenspace** of an eigenvalue λ: $E_\lambda = \{v \in \mathbb{R}^n \mid Av = \lambda v\} = \ker(A - \lambda I_n)$ (all eigenvectors for λ, plus 0).
- If some eigenvectors of A form a basis of $\mathbb{R}^n$, it is an **eigenbasis**.
- The **spectrum** (eigenspectrum) of A is the list of its eigenvalues, repeated according to multiplicity. The largest $|\lambda|$ is the **spectral radius**.

THEOREM / PROPOSITION // Eigenspaces are subspaces

$E_\lambda = \ker(A - \lambda I_n)$, hence a subspace of $\mathbb{R}^n$.

RECEIPTS (proof) //

$Av = \lambda v \iff Av - \lambda v = 0 \iff (A - \lambda I_n)v = 0$. So E_λ is the kernel of the linear map $x \mapsto (A - \lambda I_n)x$, and kernels are subspaces (Lecture 1). ■ *(and that's on periodt)*

6.2 Finding eigenvalues: this is where determinants come in

THEOREM / PROPOSITION // Eigenvalues are the roots of the characteristic polynomial

λ is an eigenvalue of $A \iff \det(A - \lambda I_n) = 0$.

RECEIPTS (proof) // a chain of equivalences

λ eigenvalue $\iff$ some $v \neq 0$ has $(A - \lambda I_n)v = 0 \iff \ker(A - \lambda I_n) \neq \{0\} \iff A - \lambda I_n$ is not invertible $\iff \det(A - \lambda I_n) = 0$. ■ *(and that's on periodt)*

EXAM CHEAT CODE // "Find the eigenvalues and eigenvectors of A"

1. Write $A - \lambda I_n$ (subtract λ on the diagonal only).
2. Compute $\chi_A(\lambda) = \det(A - \lambda I_n)$ with the Path B toolkit, keeping it **factored**.
3. Eigenvalues = roots of χ_A.
4. For each eigenvalue λ_0: solve $(A - \lambda_0 I_n)v = 0$ by Gaussian elimination. The solutions (minus 0) are the eigenvectors; a basis of the solution set is a basis of E_{λ_0}.
5. Check: $Av = \lambda_0 v$ for each vector found.

EXAMPLE // Full worked example

$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

1. $\chi_A(\lambda) = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3)$ (Section 3). Eigenvalues 3 and 1.
2. $\lambda = 3$: $A - 3I = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$. The system $-v_1 + v_2 = 0$ gives $v = (t, t)$: $E_3 = \operatorname{span}\big((1,1)\big)$. Check: $A(1,1)^\top = (3,3)^\top = 3(1,1)^\top$ ✓
3. $\lambda = 1$: $A - I = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. The system $v_1 + v_2 = 0$ gives $v = (t, -t)$: $E_1 = \operatorname{span}\big((1,-1)\big)$. Check: $A(1,-1)^\top = (1,-1)^\top$ ✓

EXAMPLE // Two teasers for next lectures

1. **No real eigenvalue:** the rotation by 90°, $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, has $\chi_R(\lambda) = \lambda^2 + 1$, no real root. Of course: a rotation by 90° keeps no direction. (Over $\mathbb{C}$: $\pm i$.)
2. **Not enough eigenvectors:** $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has $\chi_N(\lambda) = \lambda^2$, only eigenvalue 0, and $E_0 = \ker N = \operatorname{span}(e_1)$ is a line. You can't build a basis of eigenvectors: N is not diagonalisable.

6.3 Diagonalisation

THEOREM / PROPOSITION // Diagonalisation from an eigenbasis

Suppose $A \in \mathbb{R}^{n \times n}$ has n linearly independent eigenvectors $v_1, \dots, v_n$ with $Av_i = \lambda_i v_i$. Let $V = (v_1 \ \cdots \ v_n)$ (eigenvectors as columns) and $D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. Then V is invertible,

$$AV = VD \qquad \text{and} \qquad A = VDV^{-1}.$$

We say A is **diagonalisable**.

RECEIPTS (proof) // column by column

n independent vectors of $\mathbb{R}^n$ form a basis, so $\operatorname{rank} V = n$ and V is invertible. Column view: $AV = (Av_1 \ \cdots \ Av_n) = (\lambda_1 v_1 \ \cdots \ \lambda_n v_n)$. And multiplying V on the *right* by D multiplies its i-th column by λ_i: $VD = (\lambda_1 v_1 \ \cdots \ \lambda_n v_n)$. So $AV = VD$; multiply on the right by V^{-1}: $A = VDV^{-1}$. Entry-wise: $(VD)_{ij} = \sum_k V_{ik} D_{kj} = V_{ij}\lambda_j$ and $(AV)_{ij} = (Av_j)_i = \lambda_j V_{ij}$. ■ *(and that's on periodt)*

Meaning: D is the matrix of the same linear map in the eigenbasis $(v_1, \dots, v_n)$, and A is its matrix in the canonical basis. It's Lecture 2's change of basis with $P = V$: $D = V^{-1}AV$.

COMMON L (don't do this) // $VD \neq DV$

VD multiplies the **columns** of V by the λ_i; DV multiplies the **rows**. With $V = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, $D = \operatorname{diag}(3,1)$: $VD = \begin{pmatrix} 3 & 1 \\ 3 & -1 \end{pmatrix}$ but $DV = \begin{pmatrix} 3 & 3 \\ 1 & -1 \end{pmatrix}$. The correct identity is $AV = VD$;

check: $AV = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 3 & -1 \end{pmatrix} = VD$ ✓.

THEOREM / PROPOSITION // Powers of a diagonalisable matrix (classic exercise)

If $Av_i = \lambda_i v_i$ with $v_1, \ldots, v_n$ independent, then for every $k \geq 1$: $A^k v_i = \lambda_i^k v_i$, and

$$A^k = VD^kV^{-1} = V\,\mathrm{diag}(\lambda_1^k, \ldots, \lambda_n^k)V^{-1}.$$

So every A^k is diagonalisable with the same eigenvectors and eigenvalues λ_i^k. (If some $\lambda_i = 0$, then 0 is an eigenvalue of A^k; for $k = 0$, $A^0 = I_n$ with all eigenvalues 1.)

RECEIPTS (proof) // induction + domino cancellation

$A^{k+1}v_i = A(A^k v_i) = A(\lambda_i^k v_i) = \lambda_i^k Av_i = \lambda_i^{k+1} v_i$. And $A^2 = (VDV^{-1})(VDV^{-1}) = VD\underbrace{V^{-1}V}_{I}DV^{-1} = VD^2V^{-1}$; by induction $A^k = VD^kV^{-1}$, and $D^k = \mathrm{diag}(\lambda_i^k)$. ■ *(and that's on periodt)*

EXAMPLE // Powers for free

$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = VDV^{-1}$ with $D = \mathrm{diag}(3, 1)$, so $A^5 = V\,\mathrm{diag}(243, 1)V^{-1} = \begin{pmatrix} 122 & 121 \\ 121 & 122 \end{pmatrix}$ (same as in Lecture 2). And $\det(A^5) = (\det A)^5 = 3^5 = 243 = \det(D^5)$ ✓.

7 The exercises, fully solved

TL;DR // How to use this section

Try first, then check. The answers were cross-checked against reference solutions where those exist (Ex. 1, 2a, 3a, 4, the Vandermonde and block-matrix bonuses); the other exercises are solved here in full. Every final answer was also checked by computer algebra. Each solution names the Path B move it uses, so you can see the toolkit in action.

Exercise 1

SOLUTION // 1(a)

$$\begin{vmatrix} x & y & x+y \\ y & x+y & x \\ x+y & x & y \end{vmatrix}$$

Move used: "all lines sum to the same thing". Each row sums to $2(x+y)$.

1. $C_1 \leftarrow C_1 + C_2 + C_3$ (unchanged): the first column becomes $2(x+y)\,(1,1,1)^\top$. Factor it out:
$$\Delta = 2(x+y)\begin{vmatrix} 1 & y & x+y \\ 1 & x+y & x \\ 1 & x & y \end{vmatrix}.$$
2. $R_1 \leftarrow R_1 - R_3$, $R_2 \leftarrow R_2 - R_3$ (unchanged):
$$\Delta = 2(x+y)\begin{vmatrix} 0 & y-x & x \\ 0 & y & x-y \\ 1 & x & y \end{vmatrix}.$$
3. Expand along column 1: only the 1 at position $(3,1)$ survives, sign $(-1)^4 = +$:
$$\Delta = 2(x+y)\begin{vmatrix} y-x & x \\ y & x-y \end{vmatrix} = 2(x+y)\big[(y-x)(x-y) - xy\big].$$
4. $(y-x)(x-y) - xy = -(x-y)^2 - xy = -(x^2 - xy + y^2)$.
$$\boxed{\Delta = -2(x+y)(x^2 - xy + y^2) = -2(x^3 + y^3)}$$

Check with $x=1$, $y=0$: $\begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{vmatrix} = 1(0-1) - 0 + 1(0-1) = -2$, and $-2(1+0) = -2$ ✓.

SOLUTION // 1(b)

$$\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a+2)^2 & (b+2)^2 & (c+2)^2 \end{vmatrix}$$

Moves used: row operations to simplify, then recognise a Vandermonde.

1. Expand the squares: row 2 is $(a^2 + 2a + 1, \dots)$, row 3 is $(a^2 + 4a + 4, \dots)$.
2. $R_2 \leftarrow R_2 - R_1$, $R_3 \leftarrow R_3 - R_1$: rows become $(2a{+}1,\ 2b{+}1,\ 2c{+}1)$ and $(4a{+}4,\ 4b{+}4,\ 4c{+}4)$.

3. $R_3 \leftarrow R_3 - 2R_2$: row 3 becomes $(2, 2, 2)$.
4. $R_2 \leftarrow R_2 - \frac{1}{2}R_3$: row 2 becomes $(2a, 2b, 2c)$.
5. Factor 2 out of row 2 and 2 out of row 3: $\Delta = 4\begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$.
6. Swap $R_1 \leftrightarrow R_3$ (one swap: factor -1): $\Delta = -4\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = -4V(a, b, c)$.
7. Compute $V(a, b, c)$: $C_2 \leftarrow C_2 - C_1$, $C_3 \leftarrow C_3 - C_1$, expand along row 1, then factor:

$$V(a,b,c) = \begin{vmatrix} b-a & c-a \\ b^2-a^2 & c^2-a^2 \end{vmatrix} = (b-a)(c-a)\begin{vmatrix} 1 & 1 \\ b+a & c+a \end{vmatrix} = (b-a)(c-a)(c-b).$$

$$\boxed{\Delta = -4(b-a)(c-a)(c-b)}$$

Check with $(a, b, c) = (0, 1, 2)$: $\begin{vmatrix} 0 & 1 & 4 \\ 1 & 4 & 9 \\ 4 & 9 & 16 \end{vmatrix} = 0 - 1(16-36) + 4(9-16) = 20 - 28 = -8$, and $-4 \cdot 1 \cdot 2 \cdot 1 = -8$ ✓.

Exercise 2

SOLUTION // 2(a)

$$\begin{vmatrix} 1 & \cos a & \cos 2a \\ 1 & \cos b & \cos 2b \\ 1 & \cos c & \cos 2c \end{vmatrix}$$

Reminder: $\cos(2\theta) = 2\cos^2\theta - 1$.

1. Replace $\cos 2a$ by $2\cos^2 a - 1$ in column 3, then $C_3 \leftarrow C_3 + C_1$ (unchanged): column 3 becomes $(2\cos^2 a,\ 2\cos^2 b,\ 2\cos^2 c)$.
2. Factor 2 out of column 3: $\Delta_a = 2\begin{vmatrix} 1 & \cos a & \cos^2 a \\ 1 & \cos b & \cos^2 b \\ 1 & \cos c & \cos^2 c \end{vmatrix}$.
3. This is the transpose of the Vandermonde matrix of $(\cos a, \cos b, \cos c)$, and $\det M^\top = \det M$.

$$\boxed{\Delta_a = 2(\cos b - \cos a)(\cos c - \cos a)(\cos c - \cos b)}$$

SOLUTION // 2(b)

$$\begin{vmatrix} \cos(a-b) & \cos(b-c) & \cos(c-a) \\ \cos(a+b) & \cos(b+c) & \cos(c+a) \\ \sin(a+b) & \sin(b+c) & \sin(c+a) \end{vmatrix}$$

Each column is built from a pair $(u, v) \in \{(a, b), (b, c), (c, a)\}$. Write $c_u = \cos u$, $s_u = \sin u$, $t_u = \tan u$. The column for (u, v) is $\big(c_u c_v + s_u s_v,\ \ c_u c_v - s_u s_v,\ \ s_u c_v + c_u s_v\big)^\top$.

1. $R_1 \leftarrow R_1 + R_2$: row 1 becomes $2c_u c_v$. Then $R_2 \leftarrow R_2 - \frac{1}{2}R_1$: row 2 becomes $-s_u s_v$. Both

unchanged-det moves. Factor 2 from row 1 and -1 from row 2:

$$\Delta_b = -2\det N, \qquad N = \text{columns } \big(c_u c_v,\ s_u s_v,\ s_u c_v + c_u s_v\big)^\top.$$

2. Assume $\cos a, \cos b, \cos c \neq 0$. Factor $c_u c_v$ out of each column (total factor $\cos^2 a \cos^2 b \cos^2 c$): the column becomes $(1,\ t_u t_v,\ t_u + t_v)^\top$.

$$\det N = \cos^2 a \cos^2 b \cos^2 c \begin{vmatrix} 1 & 1 & 1 \\ t_a t_b & t_b t_c & t_c t_a \\ t_a + t_b & t_b + t_c & t_c + t_a \end{vmatrix}.$$

3. $C_2 \leftarrow C_2 - C_1$ gives $(0,\ t_b(t_c - t_a),\ t_c - t_a) = (t_c - t_a)(0, t_b, 1)$; $C_3 \leftarrow C_3 - C_1$ gives $(0,\ t_a(t_c - t_b),\ t_c - t_b) = (t_c - t_b)(0, t_a, 1)$. Factor both out, then expand along row 1:

$$\begin{vmatrix} 1 & 0 & 0 \\ t_a t_b & (t_c - t_a)t_b & (t_c - t_b)t_a \\ t_a + t_b & t_c - t_a & t_c - t_b \end{vmatrix}$$

$$= (t_c - t_a)(t_c - t_b) \begin{vmatrix} 1 & 0 & 0 \\ t_a t_b & t_b & t_a \\ t_a + t_b & 1 & 1 \end{vmatrix} = (t_c - t_a)(t_c - t_b) \begin{vmatrix} t_b & t_a \\ 1 & 1 \end{vmatrix} \quad \text{(expand along row 1)},$$

and $\begin{vmatrix} t_b & t_a \\ 1 & 1 \end{vmatrix} = t_b - t_a$. So the determinant is $(t_c - t_a)(t_c - t_b)(t_b - t_a)$.

4. Use $\tan p - \tan q = \dfrac{\sin(p-q)}{\cos p \cos q}$: the product of the three differences is $\dfrac{\sin(c-a)\sin(c-b)\sin(b-a)}{\cos^2 a \cos^2 b \cos^2 c}$, so $\det N = \sin(c-a)\sin(c-b)\sin(b-a)$.
5. $\Delta_b = -2\sin(c-a)\sin(c-b)\sin(b-a)$. Flipping the three differences gives a factor $(-1)^3$.
6. If some cosine is 0: both sides are continuous functions of (a, b, c) that agree wherever the cosines are non-zero (a dense set), so they agree everywhere.

$$\boxed{\Delta_b = 2\sin(a-b)\sin(b-c)\sin(a-c)}$$

Exercise 3

SOLUTION // 3(a)

$$\Delta_x = \begin{vmatrix} 1 & 1 & 2 & 3 \\ 1 & 2 - x^2 & 2 & 3 \\ 2 & 3 & 1 & 5 \\ 2 & 3 & 1 & 9 - x^2 \end{vmatrix}$$

First, guess the roots (Path B cheat code "with a parameter"):

- $x = \pm 1$: $2 - x^2 = 1$, so rows 1 and 2 are equal: $\Delta_{\pm 1} = 0$.
- $x = \pm 2$: $9 - x^2 = 5$, so rows 3 and 4 are equal: $\Delta_{\pm 2} = 0$.

So we expect the factors $(1 - x^2)$ and $(4 - x^2)$. (A non-invertible matrix has some row that is a combination of the others.)

Now compute.

1. $R_4 \leftarrow R_4 - R_3$ (using the original row 3): row 4 becomes $(0, 0, 0, 4 - x^2)$.

2. $R_2 \leftarrow R_2 - R_1$: row 2 becomes $(0,\ 1-x^2,\ 0,\ 0)$. $R_3 \leftarrow R_3 - 2R_1$: row 3 becomes $(0,\ 1,\ -3,\ -1)$.
$$\Delta_x = \begin{vmatrix} 1 & 1 & 2 & 3 \\ 0 & 1-x^2 & 0 & 0 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 4-x^2 \end{vmatrix}.$$
3. Expand along column 1 (only the top 1): $\Delta_x = \begin{vmatrix} 1-x^2 & 0 & 0 \\ 1 & -3 & -1 \\ 0 & 0 & 4-x^2 \end{vmatrix}$.
4. Expand along row 1 (only $1-x^2$): $\Delta_x = (1-x^2)\begin{vmatrix} -3 & -1 \\ 0 & 4-x^2 \end{vmatrix} = (1-x^2)(-3)(4-x^2)$.

$$\boxed{\Delta_x = -3(1-x^2)(4-x^2)}$$

Consistent with the cases $x = \pm 1$ and $x = \pm 2$ ✓.

SOLUTION // 3(b)

$$\begin{vmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1-x & 1 & 1 \\ 1 & 1 & 1+z & 1 \\ 1 & 1 & 1 & 1-z \end{vmatrix}$$

Moves used: subtract rows to isolate x and z, then columns to create zeros.

1. $R_1 \leftarrow R_1 - R_2$: row 1 becomes $(x,\ x,\ 0,\ 0)$. $R_3 \leftarrow R_3 - R_4$: row 3 becomes $(0,\ 0,\ z,\ z)$.
2. $C_2 \leftarrow C_2 - C_1$ and $C_4 \leftarrow C_4 - C_3$:
$$\Delta = \begin{vmatrix} x & 0 & 0 & 0 \\ 1 & -x & 1 & 0 \\ 0 & 0 & z & 0 \\ 1 & 0 & 1 & -z \end{vmatrix}.$$
3. Expand along row 1 (only x, sign $+$): $\Delta = x\begin{vmatrix} -x & 1 & 0 \\ 0 & z & 0 \\ 0 & 1 & -z \end{vmatrix}$.
4. Expand along column 1 (only $-x$): $\Delta = x \cdot (-x)\begin{vmatrix} z & 0 \\ 1 & -z \end{vmatrix} = -x^2 \cdot (-z^2)$.

$$\boxed{\Delta = x^2 z^2}$$

Check: $x = 0$ makes rows 1 and 2 equal, $z = 0$ makes rows 3 and 4 equal: $\Delta = 0$ in both cases ✓.

Exercise 4

$T : \mathcal{M}_n(\mathbb{R}) \to \mathcal{M}_n(\mathbb{R})$, $T(M) = M^\top$. *Find* $\det(T)$ *(as an operator on the* n^2*-dimensional space* $\mathcal{M}_n(\mathbb{R})$*).*

SOLUTION // 4 – method 1: an adapted basis (standard route)

1. T **is linear:** $(\lambda M + \mu N)^\top = \lambda M^\top + \mu N^\top$.
2. **Quick remark:** $T \circ T = \mathrm{Id}$, so $\det(T)^2 = \det(\mathrm{Id}) = 1$ and $\det T = \pm 1$. But which one?
3. Let $\mathcal{S}_n$ (symmetric, $M^\top = M$) and $\mathcal{A}_n$ (antisymmetric, $M^\top = -M$). Both are subspaces, and $\mathcal{M}_n(\mathbb{R}) = \mathcal{S}_n \oplus \mathcal{A}_n$: every $M = \frac{1}{2}(M + M^\top) + \frac{1}{2}(M - M^\top)$ (Lecture 2), and $\mathcal{S}_n \cap \mathcal{A}_n = \{0\}$ ($M = M^\top = -M$ forces $M = 0$).
4. Bases (with E_{ij} = the matrix with a 1 at (i, j), zeros elsewhere):
   - $\mathcal{S}_n$: $\{E_{ii}\}_{1 \le i \le n} \cup \{E_{ij} + E_{ji}\}_{i<j}$, so $\dim \mathcal{S}_n = n + \frac{n(n-1)}{2} = \frac{n(n+1)}{2}$;
   - $\mathcal{A}_n$: $\{E_{ij} - E_{ji}\}_{i<j}$, so $\dim \mathcal{A}_n = \frac{n(n-1)}{2}$.
5. In the concatenated basis, T fixes each symmetric basis matrix ($T(S) = S$) and negates each antisymmetric one ($T(A) = -A$). Its matrix is diagonal: $\mathrm{diag}(\underbrace{1, \dots, 1}_{n(n+1)/2}, \underbrace{-1, \dots, -1}_{n(n-1)/2})$.

$$\boxed{\det(T) = (-1)^{\frac{n(n-1)}{2}}}$$

SOLUTION // 4 – method 2: a Path A shortcut

In the canonical basis (E_{ij}), $T(E_{ij}) = E_{ji}$: T just *permutes* the basis. It fixes the n matrices E_{ii} and swaps $E_{ij} \leftrightarrow E_{ji}$ for each of the $\frac{n(n-1)}{2}$ pairs $i < j$. So its matrix is a permutation matrix of a product of $\frac{n(n-1)}{2}$ disjoint transpositions, and its determinant is the signature $(-1)^{n(n-1)/2}$. Same answer.
Examples: $n = 2$: $\det T = -1$ (one swap $E_{12} \leftrightarrow E_{21}$). $n = 3$: three swaps, $\det T = -1$. $n = 4$: six swaps, $\det T = +1$.

Exercise 5

Determine all $A \in \mathcal{M}_3(\mathbb{C})$ *such that* $\det(A + M) = \det(A) + \det(M)$ *for every* $M \in \mathcal{M}_3(\mathbb{C})$.

SOLUTION // 5

Answer: only $A = 0$**.** (The trap from Section 1: the determinant is not additive.)
$A = 0$ **works:** $\det(0 + M) = \det M = \det 0 + \det M$. ✓
Conversely, suppose A has the property.

1. $\det A = 0$**.** Take $M = A$: $\det(2A) = 2 \det A$. But $\det(2A) = 2^3 \det A = 8 \det A$. So $8 \det A = 2 \det A$, hence $\det A = 0$.
2. **So the property reads** $\det(A + M) = \det M$ **for all** M**.** Suppose $A \neq 0$ and look for a contradiction: a matrix M with $\det M = 0$ but $\det(A + M) \neq 0$.
3. Since $A \neq 0$, some column a_k of A is non-zero. Complete a_k into a basis (a_k, u, w) of $\mathbb{C}^3$ (incomplete basis theorem, Lecture 1).
4. Build M column by column: column k of M is 0; the two other columns j of M are chosen as $u - a_j$ and $w - a_j$. Then:
   - M has a zero column, so $\det M = 0$;
   - the columns of $A + M$ are a_k, u, w (in some order): a basis, so $\det(A + M) \neq 0$.
5. Contradiction with $\det(A + M) = \det M = 0$. So $A = 0$.

Exercise 6 (*)

SOLUTION // 6(a)

$$\Delta_n = \begin{vmatrix} 1 & 2 & 3 & \cdots & n \\ -1 & 0 & 3 & \cdots & n \\ -1 & -2 & 0 & \cdots & n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -1 & -2 & -3 & \cdots & 0 \end{vmatrix}$$

Entry (i, j) is j above the diagonal, $-j$ below, 0 on the diagonal except $a_{11} = 1$.

1. For every $i \geq 2$: $R_i \leftarrow R_i + R_1$ (unchanged). In row i: below the diagonal, $-j + j = 0$; on the diagonal, $0 + i = i$; above, $j + j = 2j$.
2. The matrix is now upper triangular with diagonal $1, 2, 3, \ldots, n$.

$$\boxed{\Delta_n = 1 \cdot 2 \cdot 3 \cdots n = n!}$$

Check $n = 2$: $\begin{vmatrix} 1 & 2 \\ -1 & 0 \end{vmatrix} = 0 + 2 = 2 = 2!$ ✓. $n = 3$: $\begin{vmatrix} 1 & 2 & 3 \\ -1 & 0 & 3 \\ -1 & -2 & 0 \end{vmatrix} = 6 = 3!$ ✓.

SOLUTION // 6(b)

$$\Delta_n = \begin{vmatrix} 1 & x_1 & x_2 & \cdots & x_n \\ 1 & x & x_2 & \cdots & x_n \\ 1 & x_1 & x & \cdots & x_n \\ \vdots & & & \ddots & \vdots \\ 1 & x_1 & x_2 & \cdots & x \end{vmatrix} \quad (\text{size } n+1)$$

Row $k+1$ ($k = 1, \ldots, n$) equals row 1 except that its entry in column $k+1$ is x instead of x_k.

1. For every $k \geq 1$: $R_{k+1} \leftarrow R_{k+1} - R_1$. Row $k+1$ becomes $(0, \ldots, 0,\ x - x_k,\ 0, \ldots, 0)$, with $x - x_k$ on the diagonal.
2. The matrix is upper triangular with diagonal $1,\ x - x_1,\ x - x_2,\ \ldots,\ x - x_n$.

$$\boxed{\Delta_n = \prod_{k=1}^{n} (x - x_k)}$$

Check: $x = x_k$ makes row $k+1$ equal to row 1, so $\Delta_n = 0$ ✓.

Exercise 7 – block matrices

SOLUTION // 7(1)

$$\det \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = \det(A) \det(D) \quad (A \in \mathcal{M}_p,\ D \in \mathcal{M}_q \text{ square})$$

Write $M = \begin{pmatrix} A & B \\ 0 & D \end{pmatrix}$.

Case 1: $\det A = 0$**.** Then some column of A is a combination of the other columns of A. Below A there are only zeros, so the same combination works for the first p columns of M: the columns of M are dependent, $\det M = 0 = \det A \det D$.

Case 2: $\det A \neq 0$. Then A^{-1} exists and (block multiplication, check it entry-block by entry-block)

$$M = \begin{pmatrix} A & 0 \\ 0 & I_q \end{pmatrix} \begin{pmatrix} I_p & A^{-1}B \\ 0 & D \end{pmatrix}, \qquad \text{since} \quad \begin{pmatrix} A \cdot I_p + 0 & A \cdot A^{-1}B + 0 \cdot D \\ 0 + 0 & 0 + I_q D \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & D \end{pmatrix}.$$

- $\det \begin{pmatrix} A & 0 \\ 0 & I_q \end{pmatrix} = \det A$: expand along the last column (a single 1 on the diagonal, sign +), then the new last column, q times.
- $\det \begin{pmatrix} I_p & A^{-1}B \\ 0 & D \end{pmatrix} = \det D$: expand along the first column (a single 1), then the new first column, p times.

By multiplicativity, $\det M = \det A \det D$. ■ *(and that's on periodt)*

SOLUTION // 7(2)

$$\text{If } A \text{ is invertible: } \det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(A) \det(D - CA^{-1}B)$$

Idea: block Gaussian elimination. Kill C with "$R_2 \leftarrow R_2 - CA^{-1}R_1$", which is multiplication on the left by a block lower-triangular matrix with identity blocks.

1. Check the factorisation:
$$\begin{pmatrix} I_p & 0 \\ CA^{-1} & I_q \end{pmatrix} \begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix} = \begin{pmatrix} A & B \\ CA^{-1}A & CA^{-1}B + D - CA^{-1}B \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}.$$
2. The first factor is block lower triangular; its transpose is block upper triangular with blocks I_p, I_q, so by part (1) its determinant is $1 \cdot 1 = 1$.
3. The second factor is block upper triangular: determinant $\det A \cdot \det(D - CA^{-1}B)$ by part (1).
4. Multiplicativity gives the result.

($D - CA^{-1}B$ is the **Schur complement** of A. It's everywhere in statistics: conditioning a Gaussian vector uses exactly this matrix.)

Exercise 8 (*) – Vandermonde

SOLUTION // 8(1) $\det V_n(x_1, \ldots, x_n) = \prod_{1 \leq i < j \leq n} (x_j - x_i)$

Write $V(x_1, \ldots, x_n)$ for this determinant (row i = the powers $x_1^{i-1}, \ldots, x_n^{i-1}$).
Non-distinct case. If $x_i = x_j$ for some $i \neq j$, columns i and j are equal: $V = 0$, and the product contains the factor $x_j - x_i = 0$. The formula holds.
Distinct case, by induction on n.

1. $n = 2$: $\begin{vmatrix} 1 & 1 \\ x_1 & x_2 \end{vmatrix} = x_2 - x_1$. ✓
2. **Heredity.** Assume the formula for $n - 1$. Replace x_n by a variable X: $P(X) = V(x_1, \ldots, x_{n-1}, X)$.
3. Expanding along the last column (entries $1, X, \ldots, X^{n-1}$), P is a polynomial of degree $\leq n-1$, and the coefficient of X^{n-1} is the cofactor of position (n, n), i.e. $(+1) \cdot V(x_1, \ldots, x_{n-1})$.

4. For $i \le n-1$: $P(x_i) = 0$ (two equal columns). So $x_1, \dots, x_{n-1}$ are $n-1$ distinct roots of P, and $P(X) = V(x_1, \dots, x_{n-1}) \prod_{i=1}^{n-1}(X - x_i)$.
5. Take $X = x_n$ and use the induction hypothesis:

$$V(x_1, \dots, x_n) = \prod_{1 \le i < j \le n-1} (x_j - x_i) \cdot \prod_{i=1}^{n-1} (x_n - x_i) = \prod_{1 \le i < j \le n} (x_j - x_i).$$

Assumptions on the x_i? None are needed for the *formula*: it holds for all real numbers (the non-distinct case is trivial). Distinctness is only used inside the proof. What it tells you: V_n is invertible $\iff$ the x_i are pairwise distinct.

SOLUTION // 8(2)

$$\begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \\ 1 & 4 & 16 & 64 \end{vmatrix}$$

Row k is $(1, x, x^2, x^3)$ with $x = k$: it is the *transpose* of $V_4(1, 2, 3, 4)$, and transposing doesn't change the determinant.

$$\det = (2-1)(3-1)(4-1)(3-2)(4-2)(4-3) = 1 \cdot 2 \cdot 3 \cdot 1 \cdot 2 \cdot 1 = \boxed{12}.$$

SOLUTION // 8(3) Where else does Vandermonde appear?

- **Exercise 1(b):** after row operations, $\Delta = -4V(a, b, c)$.
- **Exercise 2(a):** $\Delta_a = 2 \det V(\cos a, \cos b, \cos c)^\top$.

ML link: fitting a polynomial of degree $n-1$ through n points (x_i, y_i) means solving a linear system whose matrix is a Vandermonde matrix; it has a unique solution exactly when the x_i are distinct, i.e. when $\det V \neq 0$.

8 Final boss: the cheat sheet

8.1 "If they ask you this, do that"

If asked...	**Do this**
det of a 2×2 / 3×3	$ad-bc$ / Sarrus (or expand along a line with zeros).
det of a big matrix	Create zeros with $R_i \leftarrow R_i + \lambda R_j$ (free), then expand along the zero-rich line, or go triangular.
Minor / cofactor of a_{ij}	Delete row i and column j: M_{ij}. Minor $= \det M_{ij}$. Cofactor $= (-1)^{i+j}\det M_{ij}$ (grid sign).
Rows/columns all sum to the same thing	Add all columns into the first (or all rows into one), factor out the common value.
Entries with a parameter	Find values making two lines equal/proportional: each gives a factor. Then compute and confirm.
Show $\det A = 0$	Exhibit a dependence between rows/columns, or a non-zero x with $Ax = 0$.
Show $\det A \neq 0$	Show A invertible (e.g. triangular with non-zero diagonal after row ops), or compute it.
Is $(v_1,\dots,v_n)$ a basis?	det of the coordinate matrix $\neq 0$.
$\det(\lambda A)$, $\det(AB)$, $\det(A^{-1})$, $\det(A^\top)$, $\det(A^k)$	$\lambda^n \det A$, $\det A \det B$, $1/\det A$, $\det A$, $(\det A)^k$.
det of a linear map $f: E \to E$	Matrix of f in *any* basis (choose a clever one), take its det.
Signature of a permutation	Count inversions (= crossings); $\varepsilon = (-1)^N$.
Eigenvalues	Roots of $\chi_A(\lambda) = \det(A - \lambda I)$; keep it factored. 2×2: $\lambda^2 - \operatorname{tr} A\,\lambda + \det A$.
Eigenvectors for λ_0	Solve $(A - \lambda_0 I)v = 0$; basis of the solution set = basis of E_{λ_0}.
Diagonalise	V = eigenvectors as columns, D = eigenvalues in the same order; check $AV = VD$; then $A = VDV^{-1}$, $A^k = VD^kV^{-1}$.
Block matrix $\begin{pmatrix} A & B \\ 0 & D \end{pmatrix}$	$\det A \det D$. If A invertible: $\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det A \det(D - CA^{-1}B)$.

8.2 Formula sheet

Leibniz	$\det A = \sum_{\sigma\in\mathfrak{S}_n} \varepsilon(\sigma) \prod_{j=1}^n a_{\sigma(j),j}, \quad \varepsilon(\sigma) = (-1)^{N(\sigma)}$
2×2, 3×3	$ad-bc$; $\quad aei + bfg + cdh - ceg - bdi - afh$
Laplace	$\det A = \sum_j (-1)^{i+j} a_{ij} \det M_{ij}$ (row i) $= \sum_i (-1)^{i+j} a_{ij} \det M_{ij}$ (column j)
Operations	swap: $\times(-1)$; $\quad$ scale a line: $\times\lambda$; $\quad R_i \leftarrow R_i + \lambda R_j$: unchanged
Algebra	$\det(AB) = \det A \det B$, $\det A^\top = \det A$, $\det(\lambda A) = \lambda^n \det A$, $\det A^{-1} = 1/\det A$
Invertibility	$\det A \neq 0 \iff A$ invertible $\iff$ columns (rows) independent
Special	triangular: $\prod a_{ii}$; $\quad$ orthogonal: ± 1; $\quad$ Vandermonde: $\prod_{i<j}(x_j - x_i)$
Similarity	$\det(P^{-1}AP) = \det A$; $\quad \det_{\mathcal{B}} = \det(P)\det_{\mathcal{B}'}$
Inverse	$A^{-1} = \frac{1}{\det A}\operatorname{Com}(A)^\top$
Eigen	λ eigenvalue $\iff \det(A - \lambda I) = 0$; $\quad E_\lambda = \ker(A - \lambda I)$; $\quad A = VDV^{-1}$

8.3 Hall of Ls

COMMON L (don't do this) // Do not

1. Say "$\det A = 0$ iff a row is zero" or "iff two rows are proportional". Those are *sufficient* signs. The iff is: rows (columns) **linearly dependent**. Counterexample: $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$.
2. Write $\det(\lambda A) = \lambda \det A$. It's λ^n.
3. Write $\det(A + B) = \det A + \det B$. False (Exercise 5).
4. Use Sarrus on a 4×4.
5. Forget the $(-1)^{i+j}$ signs in a cofactor expansion, or start the grid with $-$. Top-left is always $+$.
6. Do $R_i \leftarrow \lambda R_i + \mu R_j$ and forget that it multiplies the determinant by λ.
7. Forget the -1 for every row swap.
8. Confuse the minor *matrix* M_{ij} with the minor *number* $\det M_{ij}$, or the minor with the cofactor.
9. Read $|A|$ as an absolute value. It's the determinant (it can be negative).
10. Subtract λ everywhere in $A - \lambda I$ (only on the diagonal), or expand χ_A fully before factoring.
11. Write DV instead of VD in $AV = VD$.
12. Talk about the determinant of a non-square matrix. It doesn't exist.

8.4 The whole lecture in 10 lines

TL;DR // Screenshot this

1. $\det A$ = signed volume scaling factor of $x \mapsto Ax$. $|\det|$ = scaling, sign = orientation.
2. It's the unique function of the columns that is multilinear, alternating, and equals 1 on I_n.
3. **Path A:** $\det A = \sum_\sigma \varepsilon(\sigma) \prod_j a_{\sigma(j),j}$: one entry per row and column, signed by the parity of the permutation.
4. **Path B:** cofactor expansion along any line, with the $+ - +-$ grid; create zeros first with free row operations.
5. Swap: -1. Scale a line: $\times\lambda$. Add a multiple of another line: nothing. Triangular: product of the diagonal.
6. $\det(AB) = \det A \det B$, $\det A^\top = \det A$, $\det(\lambda A) = \lambda^n \det A$, $\det A^{-1} = 1/\det A$.
7. $\det A \neq 0 \iff$ invertible $\iff$ lines independent $\iff$ basis. $\det A = 0 \iff$ lines dependent.
8. Similar matrices share their determinant, so a linear map $f : E \to E$ has a well-defined $\det f$. Orthogonal matrices have $\det = \pm 1$.
9. Eigenvalues are the roots of $\chi_A(\lambda) = \det(A - \lambda I)$; eigenvectors span $\ker(A - \lambda I)$.

10. n independent eigenvectors $\Rightarrow A = VDV^{-1}$, $A^k = VD^kV^{-1}$. ML: volumes, Gaussians ($\det \Sigma$), information loss, rotations (RoPE).

Want more practice?

Read Chapter 4 of *Mathematics for Machine Learning* (Sections 4.1 to 4.4: determinants and eigendecomposition), as recommended for the next class.

Next episode: positive (semi-)definite matrices, Gram matrices, the spectral theorem, SVD. Stay tuned, bestie.